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2004-08

**Reliability data handbook –
Universal model for reliability prediction
of electronics components, PCBs
and equipment**



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Reliability data handbook – Universal model for reliability prediction of electronics components, PCBs and equipment

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**RELIABILITY DATA HANDBOOK –
UNIVERSAL MODEL FOR RELIABILITY PREDICTION
OF ELECTRONICS COMPONENTS, PCBs AND EQUIPMENT**

FOREWORD

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IEC 62380, which is a technical report, has been prepared by IEC technical committee 47: Semiconductor devices.

The text of this standard is based on the following documents:

Enquiry draft	Report on voting
47/1705/DTR	47/1722A/RVC

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This technical report does not follow the rules for structuring international standards as given in Part 2 of the ISO/IEC Directives.

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- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

INTRODUCTION

This reliability calculation guide for electronic and optical card, is an important progress compared to older guides. Calculation models take directly into account the influence of the environment. The thermal cycling seen by cards, function of mission profiles undergone by the equipment, replace environment factor which is difficult to evaluate. These models can handle permanent working, on/off cycling and dormant applications. On the other hand, failure rate related to the component soldering, is henceforth-included in component failure rate.

RELIABILITY DATA HANDBOOK – UNIVERSAL MODEL FOR RELIABILITY PREDICTION OF ELECTRONICS COMPONENTS, PCBs AND EQUIPMENT

1 Scope

This technical report provides elements to calculate failure rate of mounted electronic components. It makes equipment reliability optimization studies easier to carry out, thanks to the introduction of influence factors.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60086 (all parts), *Primary batteries*

IEC 60099 (all parts), *Surge arresters*

IEC 60115 (all parts), *Fixed arrestors for use in electronic equipment*

IEC 60146, (all parts), *Semiconductor convertors – General requirements and line commutated convertors*

IEC 60255 ((all parts), *Electrical relays*

IEC 60269 (all parts), *Low-voltage fuses*

IEC 61951 (all parts), *Secondary cells and batteries containing alkaline or other non-alkaline electrolytes – Portable sealed rechargeable single cells*

IEC 60326 (all parts), *Printed boards*

IEC 60368 (all parts), *Piezoelectric filters of assessed quality*

IEC 60384 (all parts), *Fixed capacitors for use in electronic equipment*

IEC 60393 (all parts), *Potentiometers for use in electronic equipment*

IEC 60535, *Jet fans and regulators*

IEC 60539 (all parts), *Directly heated negative temperature coefficient thermistors*

IEC 60721-3 (all Parts 3), *Classification of environmental conditions – Part 3: Classification of groups of environmental parameters and their severities*

IEC 60738 (all parts), *Thermistors - Directly heated positive step-function temperature coefficient*

IEC 60747 (all parts) *Semiconductor devices - Discrete devices*

IEC 60747-12 (all Parts 12) *Semiconductor devices - Part 12: Optoelectronic devices*

IEC 60747-12-2, *Semiconductor devices – Part 12: Optoelectronic devices – Section 2: Blank detail specification for laser diode modules with pigtail for fibre optic systems and sub-systems*

IEC 60748 (all parts) *Semiconductor devices – Integrated circuits*

IEC 60879, *Performance and construction of electric circulating fans and regulators*

IEC 60948, *Numeric keyboard for home electronic systems (HES)*

IEC 61019 (all parts), *Surface acoustic wave (SAW) resonators*

IEC 61051 (all parts), *Varistors for use in electronic equipment*

IEC 61248 (all parts), *Transformers and inductors for use in electronic and telecommunication equipment*

IEC 61747 (all parts), *Liquid crystal and solid-state display devices*

IEC 61261 (all parts), *Piezoelectric ceramic filters for use in electronic equipment – A specification in the IEC quality assessment system for electronic components (IECQ)*

IEC 61951 (all parts), *Secondary cells and batteries containing alkaline or other non-acid electrolytes*

IEC 61951-1, *Secondary cells and batteries containing alkaline or other non-acid electrolytes – Portable sealed rechargeable single cells*

IEC 61951-2, *Secondary cells and batteries containing alkaline or other non-acid electrolytes – Nickel-metal hydride*

IEC 62007 (all parts), *Semiconductor optoelectronic devices for fibre optic system applications*

IEC 62255 (all parts), *Multicore and symmetrical pair/quad cables for broadband digital communications (high bit rate digital access telecommunication networks) - Outside plant cables*

ETS 300 019, *Environmental engineering (EE); Environmental conditions and environmental tests for telecommunications equipment*

ISO 9000:2000, *Quality management systems – Fundamentals and vocabulary*

UTE C 96-024:1990, *Modèles thermiques simplifiés des circuits intégrés monolithiques*

3 Terms and definitions

For the purposes of this technical report, the following definitions apply.

3.1 spatial

Mission profiles corresponding to the MIL-HDBK-217F "Space; flight" environment.

NOTE Only one working phase is taken into account during each orbital revolution (LEO), or earth revolution (GEO).

Table 1 – Mission profiles for spatial

Application types	$(t_{ac})_1$ °C	τ_1	τ_{on}	τ_{off}	n_1 cycles/year	ΔT_1 °C/orbit
Low earth orbit (LEO) with On/Off cycling	40	0,15	0,15	0,85	5256	$\frac{\Delta T_{jc}}{3} + 7$
Low earth orbit (LEO) permanent working	40	1	1	0	5256	3
Geostationary earth orbit (GEO) permanent working	40	1	1	0	365	8

3.2 military

Mission profiles corresponding to the MIL-HDBK-217F "Ground; mobile" environment.

NOTE Two working phases are taken into account:

Phase 1: 36 annual switch on

Phase 2: 365 days of dormant mode

Table 2 – Mission profiles for military

Application type	$(t_{ac})_1$ °C	τ_1	τ_{on}	τ_{off}	n_1 cycles/year	ΔT_1 °C/cycle	n_2 cycles/year	ΔT_2 °C/cycle
Portable Radio	26	0,01	0,01	0,99	36	$\frac{\Delta T_j}{3} + 15$	365	8

4 Conditions of use

4.1 Introductory remarks

4.1.1 Theory of reliability predictions

Calculation of a reliability prediction for non-redundant equipment is the very first step in any complete reliability study concerning that equipment, and indeed, of any study of the reliability, availability, or safety of a system.

Reliability predictions are based on numerous assumptions, all of which need to be verified (choice of component family, for example).

A reliability study of an item entails not only verifying these assumptions, but also optimizing its reliability (qualification of components and mounting processes, minimizing risk of external failure, etc).

A reliability prediction is essential, but no more so than research into the best possible reliability for least cost.

This handbook provides all the information needed to calculate electronic component and equipped printed circuit board failure rates: failures rates delivered include the influence of component mouting processes.

4.1.2 Structure of the handbook

The handbook is specifically designed as an aid to research into how to maximize equipment reliability, and to assist in the design of the equipment, by introducing various influencing factors (see also 4.3). In order to meet this objective, it is important that any reliability prediction should begin with the start of design (and then be finalised in accordance with 4.5.4). Similarly, the choice of values for the influencing factors should not be automatic.

4.1.3 Data source

The reliability data contained in the handbook is taken mainly from field data concerning electronic equipment operating in four kinds of environment:

- a) «Ground; stationary; weather protected» (in other words: equipment for stationary_use on the ground in weather protected locations, operating permanently or otherwise).

This applies mainly to telecommunications equipment and computer hardware.

- b) «Ground; stationary; non weather protected» (in other words: equipment for stationary_use on the ground in non-weather protected locations).

This relates mainly to public payphones and GSM relays.

- c) «Airborne, Inhabited, Cargo» (in other words: equipment used in a plane, benign conditions).

This relates to on board calculators civilian planes.

- d) «Ground; non stationary; moderate» (in other words: equipment for non-stationary use on the ground in moderate conditions of use).

This concerns mainly on board automotive calculators and military mobile radio.

By processing the raw data (statistical processes, results based on geographic distribution, according to equipment type, etc.), it has been possible to include various influencing factors and eliminate the main aberrant values. Other influencing factors are derived from the experience of experts (failure analyses, construction analyses, results of endurance tests).

The values adopted are those considered most probable at the present time (1992-2001).

This databook does not give any part count values, because mission profiles are needed in order to have credible values.

4.2 Assumptions adopted for TR 62380

4.2.1 Nature of data

4.2.1.1 Reliability data

The reliability data in this handbook comprises failure rates and, for some (very few) component families, life expectancy.

Failure rates are assumed to be constant either for an unlimited period of operation (general case) or for limited periods: in these particular cases the laws governing failure rates versus time have not been adopted in the interests of simplicity.

Apart from a few exceptions (see section 4.2.1.3), the wear-out period is never reached by electronic components; in the same way it is accepted, again apart from some exceptions (see section 4.2.1.2), that the added risks of failure during the first few months of operation can be disregarded.

4.2.1.2 The infant mortality period

In practice, except for a few component families, the increased risk of failure during the first months of operation can be disregarded, because of the diversity of reasons for variations or uncertainty in the failure rate. This superficially simplistic hypothesis is in fact very realistic. It is confirmed by field data concerning the operation of equipment designed very carefully, with well chosen components (based on compatibility with use) and produced by a well controlled production system, as is generally the case for the components covered by this handbook.

4.2.1.3 Wear-out period

For the vast majority of components, the -wear-out period (during which failures take on a systematic character) is far removed from the periods of use (which range from 3 to 20 years).

There are, however, two cases in which the occurrence of wear-out failures should be taken into account (the failure rate of which increases with time):

- a) For some families, if due care is not taken, the wear-out mechanisms may give rise to systematic failures after too short a period of time; metallization electromigration in active components, for example.

This risk needs to be eliminated by a good product design, and it is important to ensure this by qualification testing. In other words, it should not be taken into account for a prediction, and should be eliminated by qualification testing and by technical evaluation, which are, therefore, of critical importance.

- b) For some (few) component families, the wear-out period is relatively short. For these families, this handbook explains how to express the period for which the failure rate can be considered constant. This life expectancy is subject to influencing factors.

Such families include relays, aluminium capacitors (with non-solid electrolyte), laser diodes, optocouplers, power transistors in cyclic operation, connectors and switches and keyboards.

For these component families, it is important to ensure that the life expectancy given by the handbook is consistent with the intended use. If not, room for manoeuvring is fairly restricted: you can reduce the stresses, change the component family (or sub-family: for aluminium capacitors with non-solid electrolyte, there are several types characterized by different qualification tests).

Provision can also be made for preventive maintenance.

NOTE: As before, and in the interests of simplicity, this handbook does not give the wear-out failure mathematical model (for which the failure rate increases over time), but a period during which the rate can be considered constant (in some cases the period at 10% of the cumulative failure rate).

4.2.2 Nature of failures

4.2.2.1 Intrinsic failures

The data in this handbook covers intrinsic failures (apart from the few exceptions given in 4.2.2.2).

In practice (see section 4.1.3), the raw reliability data has been processed to eliminate non-intrinsic component failures.

4.2.2.2 Special case of non-intrinsic residual failures due to electrical overloads

There is, necessarily, a small proportion of non-intrinsic failures in the data, because it is impossible to detect all the non-intrinsic failures when they are residual.

Take, for example, the reliability of the components used in equipment located “at the heart” of a system, which is significantly better than that of the components located at the periphery (in other words connected to the external environment). It is understood that this is due to residual overloads, since the equipment is assumed adequately protected.

For the purpose of this handbook, we have therefore included an utilisation factor to take into account nonintrinsic residual failures due to the electrical environment for active components.

4.2.2.3 Other non-intrinsic failures

The other non-intrinsic failures (due to errors of design, choice, uses) are excluded from this handbook.

Errors of this kind should be avoided; hence they are not taken into for predictions. As a matter of fact, they are very largely independent of component family.

However, for some particular objectives, such as calculation of stocks of spare parts, it may be useful to include the risks of non-intrinsic residual failures due to design errors: some indications are given in section 4.4.3.

4.2.3 Large-scale integrated circuit, production date influence

Since the 90's, the reliability growth of components no longer occur, as in the 70's and the 80's; thanks to fields failures returns data collections. This is particularly true for integrated circuits, and can be attributed to: generalization of nitride based passivations, generalization of dry etching and better planarization controls. However, the integration density for integrated circuits continues to grow at the same rate as in the past, at a constant reliability figure. For this reason, and in order to takes into account the Moore law, it is necessary to know the manufacturing year to calculate the failure rate of integrated circuits.

4.3 Influencing factors

4.3.1 Component failure rate

The component failure rate depends on a number of operational and environmental factors. This is why, for each component family, the handbook gives a base failure rate value (normally a value which corresponds to the commonest internal temperature taken as a reference) multiplied by a number of influencing factors. This simplified, empirical expression takes account of the more significant influencing factors when it comes to conditions of use.

The main factors adopted are as follows:

a) Factors giving the influence of temperature (π_t , π_w)

It is now widely accepted that temperature has a moderate effect on component reliability. The effect is significant for some families (active components and aluminum capacitors with non-solid electrolyte). The models adopted are those which give the effect of temperature on the predominating failure mechanisms (which are not normally the “wear-out” mechanisms).

For semiconductors, an Arrhenius equation has been applied with activation energy of 0.3 to 0.4 electron volts.

For passive components, an Arrhenius equation has been applied with an activation energy of 0.15 to 0.4 electron volts.

Factor π_w for potentiometers gives the influence of load resistance on the temperature rise.

In the case of power dissipating components, the thermal resistance (semiconductors) or the equation giving the internal temperature as a function of ambient temperature (resistors) has been given.

b) Factors giving the influence of special stresses:

Utilization factor π_u for thyristors, Zener diodes (operating permanently powered or otherwise).

Factor π_A for Aluminum liquid electrolyte capacitors giving the effect of current pulses.

Factor π_Y for relays (operating cycle rate).

Factor π_i for connectors (current intensity).

c) Factors giving the influence of applied voltage (π_s).

The influence of applied voltage is taken into account for transistors and optocouplers (voltage applied between input and output).

4.3.2 Life expectancy

Life expectancy, when limited, is also influenced by certain factors (optocoupler operating current; temperature of aluminum capacitors with non-solid electrolyte; contact current for relays).

Life expectancy can be expressed as a number of cycles (power transistors, switches).

4.4 How to use the data

4.4.1 Calculation method

Given that the component failure rates are assumed constant, the failure rate of a non-redundant equipment can be obtained by adding together the failure rates of its individual components. In this handbook, the failure rates given for components include the effects of the mounting on a printed circuit board, the failure rate of the naked PCB or hybrid has to be added.

Clause 6 of this handbook explains the method to be used to calculate the failure rate of a printed circuit board or a hybrid.

4.4.2 Reliability prediction results

The results of a reliability prediction are many and various, and not limited to failure rate: the following information is also obtained:

- Failure rate (of component or equipment).
- Choice of technical construction for some components (choice of component family).
- Choice of conditions of use.

4.4.3 Failure rate

The failure rate can be used directly if the aim is to identify a reference base. Such is the case for many objectives described in 4.5.

However, if the aim is to obtain an accurate estimate of stocks of spare parts, the result should be updated to take account of non-intrinsic failures:

- unconfirmed failure phenomena (equipment, subsystem, identified as defective and found to be OK on repair);
- incorrect component usage, wrong choice of components for the first months of use of equipment of new design (period of improving reliability);
- incorrect maintenance, inappropriate use, human error, environmental attack;
- production process learning factor (component mounting process, etc).

The appropriate uprating factors cannot be given in this handbook: they depend on the prior experience of a company and how new the equipment production process is (for example, for unconfirmed failures, the uprating factor ranges from 10% to over 100%, depending on newness).

4.4.4 In cases where conditions are not yet known default conditions can be assumed.

According to 4.5.1, reliability prediction calculations should begin as early as possible, at the start of the equipment design phase, even if not all the applicable conditions can yet be known: in this case default values can be used provisionally, to help determine those conditions which are as yet unknown. These default values will then be gradually discarded as the definitive conditions are identified.

This method is far preferable to the simplified calculation method (for which all the values are replaced by default values, including those, which are already known).

The calculations must therefore be prepared in such a way as to enable values to be modified easily.

4.5 Uses and aims of a reliability prediction

4.5.1 Reliability prediction as an aid to equipment design

The most beneficial use of a reliability prediction is as an aid to equipment designers. In this case, the help is based on determination of the stresses and factors influencing the reliability of each component (temperature, input voltage, technical construction of the components, etc.). Predictions based on this handbook will lead the originators of a new design to choose the best conditions and the best component families, and to draw up component qualification or evaluation programmes.

If this important objective is to be met, it is essential for the reliability prediction to be begun at the very start of design, by the design originators, and then revised as required. The work should be carried out in close collaboration with the company's component quality experts.

4.5.2 Reliability prediction to assess the potential of new equipment

The predicted reliability can be compared with the reliability objectives or stated requirements.

4.5.3 Predicted reliability values as a basis for contractual reliability values

The contractual value of a failure rate must be determined on the basis of the predicted value; these two values will not necessarily be equal: a number of contractual values may be assumed depending on observation period or certain data may be modified provided it is justified. However, in all cases, the predicted value should be taken as the base.

4.5.4 Where used in conjunction with other characteristics of a project (electrical characteristics, weight, etc.), the results of a reliability prediction can be used to compare different project solutions, such as when evaluating proposals from tenderers. Comparisons of this kind are possible only if the data used is the same, hence the existence of a reliability data handbook.

4.5.5 The predicted failure rates for the individual items of a system are crucial when calculating system dependability and reparability.

4.5.6 Reliability predictions can be used as a basis for evaluating stocks of equipment and spare components required for maintenance (however, in this case, it is important to take account of probable non-intrinsic failures, as was explained in 4.4.3). The purpose of a study of this kind is to optimize stocks of spare parts (avoid stock outages, but also avoid excessive and costly stocking levels).

4.5.7 Reliability predictions can be used as a benchmark for assessing results observed in operation. Indeed, observed results cannot be assessed effectively without a benchmark: mediocre reliability would be considered normal and there would be no attempt at improvement.

Obviously we should not expect observations to mirror exactly the predicted reliability values, for a number of reasons:

- Predictions are based only on intrinsic reliability; they do not therefore take account of external overload conditions (however, according to 4.2.2, they do take account of residual overloads).
- Predictions do not take account of design errors or incorrect use of components.
- Predictions do not take account of the risks involved in using lots of components with poor reliability.

These departures from reality, far from being a handicap are in fact an advantage; in practice, the differences can be used to reveal a lack of reliability and, following analysis, take corrective action. This very important quality enhancement process is crucial when it comes to minimizing the infant mortality period and correcting equipment design errors.

5 Environment influence

5.1 General remarks

Experience has shown that component reliability is heavily influenced by mechanical and climatic environment conditions, as well as by electrical environment conditions (residual overload).

This factor is therefore included in this handbook, based on observations and published values; for simplicity, climatic and mechanical environment conditions have been classified in ten or so environment types. However, the mission profile has to be taken into account (see 5.7), to determine estimated failure rate of components in the considered environment.

5.2 Environment types defined

The environment types are based on IEC 60721-3 («classification of groups of environmental parameters and their severity»), with some simplifications, and the specification ETS 300 019 (ETSI specification: environmental conditions for telecommunications equipment).

Table 3 gives, for the various types of environment adopted for the purposes of this handbook, the following information:

- the short form designation adopted for this handbook;
- the complete designation (generally according to IEC 60721-3);
- the main stresses included;
- some typical applications.

Table 4 quantifies the mechanical stresses (shock and vibration) for the main types of environment.

Tables 5 define the environmental conditions according to the presence and activity of chemical and mechanical substances (definitions given in table 7 based on the conventions summarized in Tables 5 and 6), and according to climatic conditions.

Table 3 – Description and typical applications of the commonest types of environment

Environment description			Description of the environment	Applications
Short form designation (adopted in the handbook)	Complete designation			
Ground; stationary; weather protected	Equipment for <u>stationary</u> use on the <u>ground</u> in <u>Weather protected locations</u>		Controlled temperature and humidity, low stress good maintenance	Equipment in environmentally controlled premises
Ground; stationary non weather protected	Equipment for stationary use on the ground; in non Weather protected locations		Some mechanical and climatic stresses (moderate) Average quality maintenance	Equipment located in premises with little or no environmental control: - phone booths - equipment in public buildings - equipment in streets, stations, etc, - equipment in industrial environments.
	important note: the phrase "non weather protected" (according to IEC 60721-3) applies to the equipment and not to the components. With regard to, the components (which are protected from the elements), the main difference from the type "ground; fixed; protected" lies in the absence of environmental control (humidity and temperature).			
Ground; non stationary; benign	Equipment for <u>non-stationary</u> use on the <u>ground</u> , in <u>benign</u> conditions		Mechanical stress is more severe than for "ground; stationary; non Weather protected" Sometimes difficult maintenance	Radiotelephones - Portable equipment on ground vehicles. Railway rolling Stock equipment
Ground; non stationary; severe	Equipment for <u>non-stationary</u> use on the <u>ground</u> , in <u>severe</u> conditions		As for "ground; non stationary benign", but with more severe; mechanical stresses	
Satellite; flight	Used on board an orbiting satellite		Very low mechanical stresses	
Satellite; launch	Used on board a satellite During launch		Extremely severe shock High amplitude vibration and high frequencies (up to 2 000 Hz)	
Airborne; benign	Used in an aircraft in ...	benign conditions	Conditions similar to those of "ground; non-stationary; benign-", but with more intense vibration up to 2 000 Hz	Other applications (other than "aircraft" and "ship") are possible, rovided that the stresses are comparable.
Airborne; moderate		moderate conditions	The qualifying terms "moderate", "severe" and "extremely severe", are defined in table 2; they represent increasing levels of mechanical stresses.	
Airborne; severe		severe conditions		
Airborne; extremely severe		extremely severe conditions		
Naval; benign	Used on board a ship in ...	benign conditions	Conditions similar to those of "ground; stationary; non-weather protected", but with more pronounced shock and vibration. The qualifying terms, "benign" and "severe" represent the mechanical stresses according to table 2.	
Naval; severe		severe conditions		

Table 4 – Mechanical conditions according to the environment: characteristic shocks and vibrations.

VIBRATIONS											SHOCKS	
Accelerations m/s^2 ↓	Frequencies Some Hz up to the above frequency. (Hz) ↓	50	100	200	200	300	300	500	1000	2000	peak acceleration m/s^2 ←	Duration ms ←
1	200	Ground stationary Weather-protected										
10	200			Ground stationary Non weather protected								
20	200		← Naval ; benign →									
20	500		← Ground ; non stationary ; benign →									
20	2000		Airborne ; benign									
30	500						← Ground; non stationary; severe →					
30	2000		Airborne; moderate									
50	200		← Naval severe →									
80	2000				Airborne; severe							
150	2000				Airborne; extremely severe						Satellite launch	

Tables 5 and 6: Represent the definition of concentration classes used in Table 7 for active substances

Table 5 – Mechanically active substances

Designation of classes used in Table 7	Sand (Mg/M3)	Dust		Examples of type of environment
		(Mg/M3)	(Mg/M2 h)	
Negligible	0	0,01	0,4	Naval; benign
Low	30	0,2	1,5	Ground; weather protected
Moderate	300	0,4	1,5	Ground; non weather protected
High	3000	4	40	Ground; non stationary; severe

Table 6 – Chemically active substances

Designation of classes used in Table 7	Salt mist	S0 ₂	H ₂ S	Cl	N0 ₂	Examples of type of environment
		(proportion in 10 ⁻⁹)				
Low	(low)*	30	7	7	50	Ground; weather protected
Moderate	(moderate)*	100	70	70	300	Ground; non weather protected
High	(high)*	400	400	70	500	Naval; severe
* No figure has been published						

Table 7 – Typical conditions for each environment type according to Table 3 (mechanically and chemically active substances and climatic conditions)

	Active substances concentration (classes according to Tables 5 and 6)			Relative humidity %	Mean temperature °C	Rapid changes of Temperature: qualitative estimation of temperature range
	Mechanically active substances	Chemically active substances				
		Gaseous substances	Fluid substances			
		Concentration class 3ccording to Table 5	Concentration class according to Table 6			
Ground; stationary; weather protected	low	low	negligible	40 to 70	+5 to +45	negligible
Ground; stationary; non weather protected	moderate	moderate	negligible	5 to 100	-40 to +45	low
Ground; non stationary; benign	moderate	moderate	negligible	5 to 100*	-40 to +45	low
Airborne; benign	low	low	negligible	5to 100	-40 to +45	low
Airborne; moderate	low	low	negligible	5 to 100	-40 to +45	low
Naval; benign	very low	low	negligible	5 to 100	-40 to +45	low
Ground; non stationary; severe	high	moderate	low	5 to 100	-40 to +70	moderate
Airborne; severe	high	moderate	high	5to-100	-65 to +85	high
Naval; severe	moderate	high	high	10 to 100	-40 to +70	high
Airborne; extremely severe	high	moderate	high	5 to 100	-65 to +85	high
Satellite; launch	low	moderate	low	0 to 50	-40 to +20	high
Satellite ; Orbit	very low	moderate	low	0	-40 à +65	moderate
* 40 to 70 on board trains (railway equipment)						

5.3 Electrical environment conditions

Reliability is also heavily dependent on electrical environment conditions (voltage and current overloads). This applies in particular to a component connected to interface circuits between an electronic circuit board and the outside environment (another equipment, especially if remotely located).

First priority is to protect the exposed components appropriately (by a system of protection comprising components designed to resist overload conditions). However, it is often found that the reliability of exposed and protected components does not match that of components located "at the heart" of an equipment. Electrical environment conditions for the active components have therefore been included (bearing in mind that the effect of residual overloads after a protection system is of concern).

The influence of the electrical environment for other families (some passive components), might equally be applied.

5.4 Validity model according to environment

Failures analysis undertaken on field failed active devices, during the period 1992 to 2001, have shown that:

- For the "ground; stationary; weather protected" environment, there is no package related defects, and nothing coming from the mounting process.
- For "ground; stationary; non-weather protected", "ground non-stationary; severe" and "airborne benign" environments, the main observed defects are caused by thermomechanical constraints applied to components mounted on PCBs. The failure rate related to the humidity is insignificant (for active components, especially since the generalization of the nitride based passivations). Furthermore, in these studied environments no defect related to mechanical shocks or to vibrations to chemical contamination has been observed. Consequently, these failure mechanisms have not been taken into account in the models.

Therefore, to use these models correctly, it is necessary to make appropriate qualification tests to verify these hypotheses for the considered environment. Plastic encapsulated devices are, in most of the described environments in this report, insensitive to shock and vibration.

Furthermore, for the "ground; stationary; non weather protected", it is necessary to ensure that there is no condensation on cold parts of the equipment (especially for equipment having a standby mode), and also there is no streaming on the equipment itself, this, to avoid any corrosion phenomenon.

5.5 Components choice

It is the responsibility of the manufacturer to guarantee the life duration specified by the final user and that components used in equipment are compatible with the environment. Therefore, premature usury phenomena shall not occur, during the useful life period of the equipment in normal utilization conditions prescribed by the final user (see 4.2.1.3).

However some components may have limited life duration, but a preventive maintenance has to be nevertheless indicated to the final user (see 4.2.1.3).

It is the responsibility of the component manufacturer to provide qualification and evaluation results of degradation mechanisms to the manufacturer and to insure that the appearance of usury mechanisms will be postponed beyond the useful life period of the equipment in normal utilization conditions, as prescribed by the final user.

Consequently, the equipment manufacturer has to choose components manufacturers who have the best "commercial practice" concerning quality, those who are ISO 9000 certified, practice the statistical process control and are under qualified manufacture line approval (or able to be).

In these conditions, there are no longer any reasons to take into consideration quality factors, and the infant mortality period related to new component technology is neglected only qualified productions lines and stabilized ones are considered here.

When an equipment manufacturer uses a new component technology, and when such a manufacturer has not been able to justify the life duration in normal use conditions of its device, the equipment manufacturer has to undertake tests allowing justification of the life duration of this component to the final user.

5.6 Learning during the deployment phase of new equipment

Models retained in this report allow for calculation of an electronic card to reach a reliability objective in its stabilised production phase. However, the operational reliability follow up of a newly developed electronic card, function of its deployment in the field, shows that there is a more or less long learning period, according to the improvement of the components implementation on the PCB and the components choice rectification for those having problem in the field (see Figure 1).

Each manufacturer has to calibrate the learning period according to his own experience. However experimentally, on many electronic cards and with several manufacturers, the ratio between the failure rate during the starting period of deployment and the one in the stabilized period, is between 2 and 3.

Consequently, as soon as the observed failure rate (out of non-defective removed cards: NDF) during the beginning of the deployment of an electronic card exceeds three times the estimated calculated value, a corrective action has to be taken.

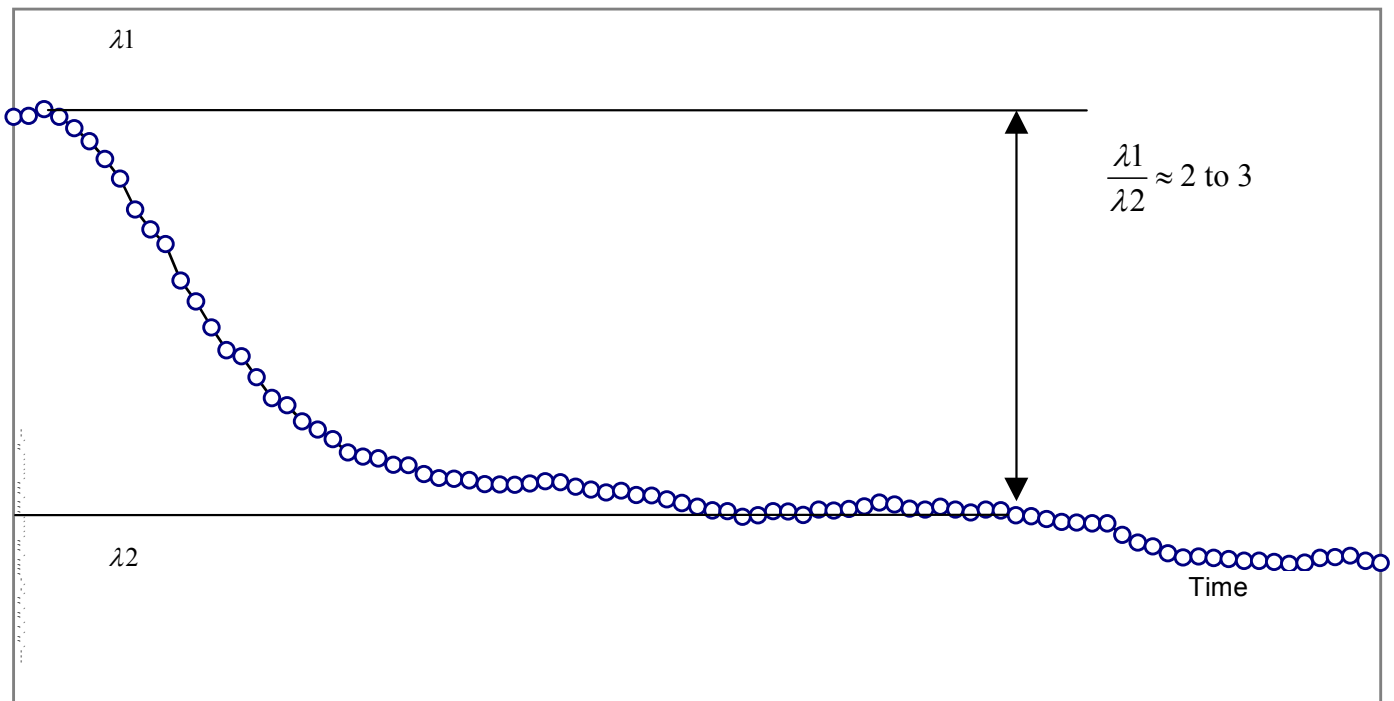


Figure 1 – Time-dependant failure rate of a new electronic printed circuit board

5.7 Mission profile

Estimated reliability calculation of equipment has to be done according to its field use conditions. They are defined by the mission profile.

A mission profile has to be decomposed in several homogeneous working phases, on the basis of a typical year of use. The following phases are to be considered:

- on/off working phases with various average outside temperatures seen by the equipment ;
- permanent-working phases with various average outside temperature swings seen by the equipment ;
- storage or dormant phases mode with various average outside temperature swings seen by the equipment.

For a reliability calculation, the time quantity which has to be taken into account on a field return coming from an equipment population, is therefore, the number of calendar hours of the installed population of this equipment, including working as well as storage or dormant hours.

Parameters necessary to define the mission profile of equipment are the following:

- $(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
- $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled (or the one of the component considered as the most critical for reliability, during the i^{th} phase of the mission profile).
- τ_i : annual ratio of times for the PCB, in permanent working mode with supply, and at the $(t_{ac})_i$ temperature.
- τ_{on} : total annual ratio of time for the PCB, in permanent working mode with supply ($\tau_{on} = \sum_{i=1}^y \tau_i$).
- τ_{off} : total annual ratio of time for the PCB, in non working or storage/dormant modes. ($\tau_{on} + \tau_{off} = 1$).
- n_i : annual number of thermal cycles seen by the components of the PCB, corresponding to the i^{th} phase of the mission profile with an average swing ΔT_i .
- ΔT_i : average swing of the thermal variation seen by the components of the PCB, corresponding to the i^{th} phase of the mission profile.

For an on/off phase the following relation exists: $\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ac})_i \right] - (t_{ae})_i$

With ΔT_j : increase of the internal temperature of the component as compared to t_{ac} , during a τ_{on} phase. (This is the junction temperature increase for an integrated circuit or a discrete device; this is the surface temperature increase for a passive device.) Only the third of its value has to be taken into account for a ΔT_i calculation, taking into account the fact that thermomechanical stresses induce defects at the solder joint of the components, but also at the wire bounding of the die. The temperature to be taken into account is therefore a compromise on the internal temperature increase of the component. Some thermal simulations have shown that a third of this value is a good compromise.

$(t_{ae})_i$: for the French climate, 11 °C is used for "Ground; stationary; non weather protected" ("ground; fixed" of MIL-HDBK-217F) environment, and 14 °C for the world-wide climate.

$(t_{ac})_i$: is obtained, taking the mean value of the temperature increase observed on the PCB near the components as compared to the external temperature of the equipment, and adding the value of $(t_{ae})_i$ for the considered phase.

t_{ac} = average temperature increase of the PCB near components + t_{ae}

For a storage or permanent working phase: ΔT_i = average of the difference between maximal and minimal temperatures per cycle seen by the equipment on the considered phase. If this value is below 3 °C, the value becomes $\Delta T_i = 0$, taking into account the fact that for these conditions, thermomechanical stresses are thermally independent in the COFFIN-MANSON equation.

For the majority of applications, one day corresponds to one cycle, and ΔT_i corresponds to the annual daily mean of the daylight / night temperature difference seen by the equipment park in the considered climate. For the French climate, $\Delta T_i = 8$ °C. For the world-wide climate, $\Delta T_i = 10$ °C.

A daily temperature variation is always superimposed on a permanent working phase according to the climatical environment of the equipment. For on/off working this daily variation is also applied on the equipment, however, only the greater temperature variation has to be taken into account, because the highest one has the main effect on the reliability of the device packages and on the mounting process.

Table 8 – Table of climates

Climate type	t_{ac} night	t_{ac} day-light	t_{ac} mean day-light/night	ΔT_i day-light/night
World-wide	5 °C	15 °C	14 °C	10 °C
France	6 °C	14 °C	11 °C	8 °C

5.8 Mission profile examples

Mission profiles described here in after are given as examples.

5.8.1 Telecoms

There is only one annual working phase to consider for a permanent working.

Table 9 is given for a permanent working. Values for "ground; stationary; non weather protected"(Ground; fixed for Mil-HDBK-217F) are given for the French climate, but other climates can be calculated.

Table 9 – Mission profiles for Telecom

Environment types	Equipment types	$(t_{ae})_i$ °C	$(t_{ac})_i$ °C	τ_1	τ_{on}	τ_{off}	n_1 cycles/year	ΔT_1 °C/cycle
Ground; benign: (G _B)	switching	20	30	1	1	0	365	0
Ground; benign: (G _B)	Transmitting	20	40	1	1	0	365	0
Ground; fixed: (G _F)	Transmitting and access	11	31	1	1	0	365	8

5.8.2 Military and civilian avionics

Mission profiles described hereinafter correspond to the MIL-HDBK-217F "Airborne; Inhabited; Cargo" environment.

Several working phases are considered.

- The working rate considers only one internal working temperature for the equipment, and takes into account the total hours of annual working.

- Three phases of thermal cycling are taken in account:

. Phase 1: first daily switch on;

- . Phase 2: switch-off between two flights, while air conditioning of the plane is working;
- . Phase 3: plane on the ground, not working.

For more complex mission profiles, all the temperature's gradient seen by components during the various different working and storage cycles have to be taken into account.

Table 10 – Mission profiles for military and civil avionics

Mission profile phases		Annual working rate for the equipment			First daily switching on		Switch-off Between two fights		Ground Non-working	
Plane types	$(t_{ac})_1$ °C	τ_1	τ_{on}	τ_{off}	n_1 cycles/year	ΔT_1 °C/cycle	n_2 cycles/year	ΔT_2 °C/cycle	n_3 cycles/year	ΔT_3 °C/cycle
A340	40	0.61	0.61	0.39	330	$\frac{\Delta T_j}{3} + 30$	330	$\frac{\Delta T_j}{3} + 15$	35	10
A330	40	0.54	0.54	0.46	330	$\frac{\Delta T_j}{3} + 30$	660	$\frac{\Delta T_j}{3} + 15$	35	10
A320	40	0.58	0.58	0.42	330	$\frac{\Delta T_j}{3} + 30$	1155	$\frac{\Delta T_j}{3} + 15$	35	10
Regional plane	40	0.61	0.61	0.39	330	$\frac{\Delta T_j}{3} + 30$	2970	$\frac{\Delta T_j}{3} + 15$	35	10
Business plane	40	0.22	0.22	0.78	300	$\frac{\Delta T_j}{3} + 30$	300	$\frac{\Delta T_j}{3} + 30$	65	10
Weapons plane	60	0.05	0.05	0.95	200	$\frac{\Delta T_j}{3} + 50$	0	0	165	10
Military cargo	50	0.05	0.05	0.95	250	$\frac{\Delta T_j}{3} + 40$	0	0	115	10
Patroller	50	0.09	0.09	0.91	300	$\frac{\Delta T_j}{3} + 40$	0	0	65	10
Helicopter	50	0.06	0.06	0.94	300	$\frac{\Delta T_j}{3} + 40$	0	0	65	10

5.8.3 Automotive

Mission profiles described hereinafter correspond to the MIL-HDBK-217F "Ground; mobile" environment.

Several working phases are considered..

- The working rates consider three different internal working temperatures for the equipment, and take into account the annual working hours for each of these temperatures. The overall working time is estimated to be 500 h.
- Two thermal cycling are considered:
Phase 1: 2 night starts;
Phase 2: 4 day light starts.
- Phase 3: non-used vehicle, dormant mode 30 days per year.

Table 11 – Mission profiles for automotive

Mission profile phases	Temp. 1		Temp. 2		Temp. 3		Ratios on/off		2 night starts		4 day light starts		Non used vehicle	
Application types	$(t_{ac})_1$ °C	τ_1	$(t_{ac})_2$ °C	τ_2	$(t_{ac})_3$ °C	τ_3	τ_{on}	τ_{off}	n_1 cycles/year	ΔT_1 °C/cycle	n_2 cycles/year	ΔT_2 °C/cycle	n_3 cycles/year	ΔT_3 °C/cycle
Motor control	32	0.020	60	0.015	85	0.023	0.058	0.942	670	$\frac{\Delta T_j}{3} + 55$	1340	$\frac{\Delta T_j}{3} + 45$	30	10
Passenger compartment	27	0.006	30	0.046	85	0.006	0.058	0.942	670	$\frac{\Delta T_j}{3} + 30$	1340	$\frac{\Delta T_j}{3} + 20$	30	10

6 Equipped printed circuit boards and hybrid circuits (IEC 60326)

6.1 Failure rate calculation of an equipped printed circuit board

Equipped board failure rate: **$(A+B) \times 10^{-9}/\text{hour}$** , with: A = connections and components ; B = board

$$A = \left[\sum \lambda_s + \sum \lambda_f + \left(1 + 3.10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times \sum \lambda_d \right]$$

Surface mounted components + Through hole components + Miscellaneous connections

λ_s : Failure rate of each particular surface mounted component (with its influence factors) expressed in $10^{-9}/\text{hour}^*$.

λ_f : Failure rate of each trough hole component (with its influence factors) expressed in $10^{-9}/\text{hour}^*$.

* If the failure rate is $3.10^{-9}/\text{h}$, take λ_s (or λ_f) =3

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i =average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.	

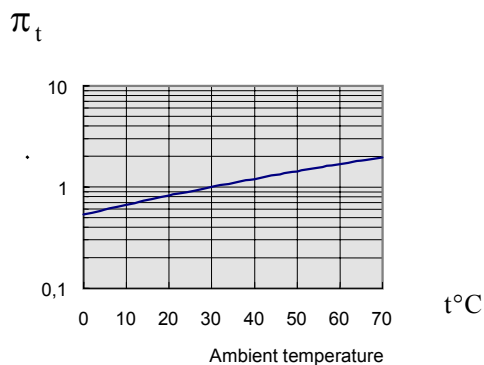
Miscellaneous connections	λ_d FIT
Manual soldering	0.5
Connecting with insulating transfer	0.5
Crimp...	0.3
Wrapped connection	0.01
Pressfit connection	0.006

$(t_{ae})_i$: average external ambient temperature of the equipment, during the i^{th} phase of the mission profile
 $(t_{ac})_i$: average internal ambient temperature, near the components, where the temperature gradient is cancelled.

$(\pi_n)_i$: i^{th} influence factor related to the annual cycle number of thermal variation, seen by the board with an amplitude of ΔT_i .

ΔT_i : i^{th} thermal variation amplitude of the mission profile.

$$B = 5.10^{-3} \cdot \pi_t \cdot \pi_c \left[N_t \sqrt{1 + \frac{N_t}{S}} + N_p \cdot \frac{1 + 0.1\sqrt{S}}{3} \cdot \pi_L \right] \times \left(1 + 3.10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right)$$



Mathematical expression for π_t
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

Number of layers influence	
Number of layers	π_C
≤ 2	1
> 2	$0.7\sqrt{\text{(number of layers)}}$

N_t = Total number of holes
(for through holes components and vias)

S = Board surface (cm^2)

N_p = number of tracks

Default value:

$$N_p = \frac{(\text{total number of connections})}{2} = \frac{\sum N_s + \sum N_f}{2}$$

N_s : number of connections for each particular surface mounted component.

N_f : number of connections for each particular through hole component.

Influence of the track width						
Predominant track width (mm)	0.56	0.35	0.23	0.15	0.10	0.08
π_L	1	2	3	4	5	6

6.2 Hybrid circuits

Hybrid circuit failure rate: $(A + B) \times 10^{-9} / \text{hour}$

components

A = Add on components and packages; B = substrate and deposited components

$$A = \sum \lambda_s + \left(0.023 \times (|\alpha_s - \alpha_c|)^{1.68} \right) \times \left(2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 0.2 \times \pi_p \times D^{1.57}$$

Add on components + package

λ_s : **Failure rate** of each add on component expressed in 10^{-9} /hour, (with its influence factors)*

* If the failure rate is $3.10^{-9} / h$, take $\lambda_s = 3$

D: hybrid circuit diagonal, or distance between forest pins, in millimeters.

α_s	Linear thermal expansion coefficient of the mounting substrate of the hybrid in ppm/°C
α_c	Linear thermal expansion coefficient of the hybrid substrate in ppm/°C

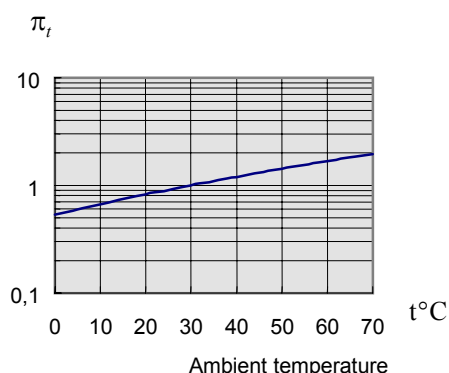
Connecting type	π_p
Single in line	1
Double in line	2
Peripheral	4

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase		$\Delta T_i = (t_{ac})_i - (t_{ae})_i$
For a permanent working phase, storage or dormant		ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.

$$B = \left\{ 5 \times 10^{-3} \pi_c \left[N_t \sqrt{1 + \frac{N_t}{S}} + N_p \frac{1 + 0.1\sqrt{S}}{3} \pi_L + 0.8 N_x \right] + \pi_i \left[\sum (0.01 R_e + 0.04 R_m) \pi_i + 0.1 C \right] \right\} \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right)$$

interconnections, crossovers

deposited components



Mathematical expression for π_t

$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273 + t_A} \right)} \text{ with } t_A : \text{ambient temperature}$$

Number of layers influence	
Number of layers	π_c
≤ 2	1
> 2	$0.7 \sqrt{(\text{number of layers})}$

S = Substrate surface (cm^2)
 N_t = number of holes for interconnections

C : Number of deposited capacitors
 R_e : Number of thick film resistors having a same factor π_i *
 R_m : Number of thin film resistors having a same factor π_i *

Precision factor *	π_i
Tolerance > 5%	1
Tolerance from 1 to 5%	1.5
Tolerance < 1%	2

*: Count apart resistors according to π_i

N_p = number of tracks
 Default value:
 $= \frac{(\text{number of components connections})}{2}$

N_x : number of crossover

Tracks width influence						
Predominant width (mm)	0,56	0,35	0,23	0,15	0,10	0,08
π_L	1	2	3	4	5	6

7 Integrated circuits

7.1 Validity domain

The end of life period of an integrated circuit is supposed to appear far beyond the utilization period of the equipment. This assumption has to be assessed by a preliminary qualification.

The main failure mechanisms to assess are following:

- For silicon technologies:
 - electromigration;
 - oxides ageing;
 - hot electrons;
 - charge gain and charge (for the write – erase cycles of the various programmable memories).
- For GaAs technologies:
 - gate sink;
 - Ohmic contact degradation;
 - gate and drain lagging;
 - electromigration.
- Packages:
 - thermal fatigue;
 - purple plague.

Estimated failure rates are valid, only if the coldest part temperature of the considered card is over the dew point temperature.

The model does not include the failure rate due to soft errors¹ provoked by the creation of electron-hole pair, on the passage of alpha particle emitted by the package materials. This failure rate due to soft errors may be of the same order of magnitude as the intrinsic failure rate, especially for dynamic memories.

For interface circuits, the models include a failure rate considered to be constant, due to external electrical influences. This failure rate depends on the electrical environment of the equipment, given that the equipment does have primary or secondary protections, depending on the state of the art of the period observed. (For the purposes of this report, interface circuits are taken to be circuits or devices connecting the equipment to the outside environment.)

7.2 Junction temperature evaluation of an integrated circuit

By default, the following simplified method will be used:

7.2.1 The junction temperature is given according to the average power dissipated in the integrated circuit by one of the following equations:

$$\begin{aligned} t_j &= t_a + P \times R_{ja} \\ t_j &= t_c + P \times R_{jc} \end{aligned}$$

in which

- t_a is the ambient temperature around the integrated circuit (°C);
- t_c is the case temperature of the integrated circuit (°C);
- t_j is the junction temperature of the integrated circuit (°C);
- R_{ja} is the junction-ambient thermal resistance of the integrated circuit (°C/W);
- R_{jc} is the junction-case thermal resistance of the integrated circuit (°C/W);
- P is the average power dissipated by the integrated circuit (Watt).

¹ When the function is fully retrievable without outside intervention.

NOTE The following equation applies:

$$\text{Junction-ambient thermal resistance} = \text{junction-case thermal resistance} + \text{case-ambient thermal resistance}$$

7.2.2 Evaluating thermal resistance

7.2.2.1 Preferred method

The preferred method is to take the thermal resistance value specified or published by the manufacturers.

7.2.2.2 Default method

By default, the values given in table 12 will be taken according to S and K, where:

S: is the number of pins of the package.

K: is the cooling factor given, according to the velocity of air V in m/s, by the following equation:

$$K = \frac{0.59V + 1.11}{V + 0.7}$$

Practical values of K are given in Table 13.

Table 12 – Thermal resistance as a function of package type, the pin number and airflow factor

	Package thermal resistance of integrated circuits as a function of: S: number of pins K: airflow factor (<i>see table 13</i>)	
	Junction-case thermal resistance R_{jc} (°C/W)	Junction-ambient thermal resistance R_{ja} (°C/W)
DIL ceramic package	$0.23 \left(10 + \frac{1520}{S+3} \right)$	$(0.23 + 0.66K) \left(10 + \frac{1520}{S+3} \right)$
DIL plastic package	$0.33 \left(10 + \frac{1520}{S+3} \right)$	$(0.23 + 0.66K) \left(10 + \frac{1520}{S+3} \right)$
PLCC plastic package	$0.28 \left(15 + \frac{1600}{S+3} \right)$	$(0.28 + 0.72K) \left(10 + \frac{1600}{S+3} \right)$
SOJ and SOL plastic package	$0.28 \left(15 + \frac{1760}{S+3} \right)$	$(0.28 + 0.72K) \left(10 + \frac{1760}{S+3} \right)$
TSOP plastic package	$0.4 \left(20 + \frac{2500}{S+3} \right)$	$(0.4 + 0.6K) \left(20 + \frac{2500}{S+3} \right)$
PGA ceramic package	$0.33 \left(10 + \frac{1440}{S+3} \right)$	$(0.33 + 0.66K) \left(10 + \frac{1440}{S+3} \right)$
QFP plastic package	$0.4 \left(27 + \frac{2260}{S+3} \right)$	$(0.4 + 0.6K) \left(27 + \frac{2260}{S+3} \right)$
BGA plastic package	$0.4 \left(6.6 + \frac{1.1 \times 10^6}{S^2} \right)$	$(0.4 + 0.6K) \left(6.6 + \frac{1.1 \times 10^6}{S^2} \right)$

Table 13 – Typical values of the air flow speed V, and the air flow factor K

	V (m/s)	K
Natural convection	0.15	1.4
Slightly assisted cooling	0.5	1.2
Fan assisted cooling	1	1
Forced cooling	4	0.7

7.2.2.3 Evaluating the average power dissipated by an integrated circuit P (Watt)

Preferably use the real average power dissipated by the integrated circuit. By default, use the following method:

7.2.2.3.1 CMOS family**7.2.2.3.1.1 CMOS digital circuit, other than memory and 74 ACT family circuits**

Calculate:

$$P = \frac{P_1(P_1 + 3.5)}{3P_1 + 5}$$

The value of P_1 is calculated from one or the other following formulas:

a) If an equivalent capacitance C_{pd} is specified :

$$P_1 = (V_{cc}^2 \times f) \times [C_{pd} (\text{number of individual functions}) + C_L (\text{number of output pin})] \times 10^{-6}$$

NOTE 2

b) If a current consumption I_s is specified at a specified frequency f_s (MHz):

$$P_1 = V_{cc} I_s \frac{f}{f_s} + V_{cc}^2 f C_L (\text{number of outputs}) \times 10^{-6}$$

where

P_1 is the maximum power (W);

V_{cc} is the supply voltage (V);

f is the working frequency (MHz);

C_{pd} is the equivalent capacitance for calculating dissipated power for each individual function, (pF);

C_L is the load capacitance on outputs, for each output, (pF);

I_s is the specified value of the consumption current at a specified frequency f_s , and when outputs are not charged;

f_s is the value of the specified frequency for the consumption (MHz).

NOTE 1 - In case b, the number of individual functions has disappeared ,because the consumption current is only specified for circuits with one complex elementary function.

NOTE 2 - Sometimes the working frequency f is assimilated to f_s .

The default values for V_{cc} , C_L , C_{pd} , f , are as follows:

V_{cc} is the 5 V or 3 V

C_L is the 50 pF

f is the 30 MHz for HC family; 50 MHz for AC family.

C_{pd} according to specification or catalogue.

7.2.2.3.1.2 CMOS digital circuits of the 74 ACT family

Calculate:

$$P = \frac{P_1(P_1 + 3.5)}{3P_1 + 5}$$

with

$$P_1 = 1,6 \cdot 10^{-3} \times V_{cc} \times (\text{number of input pins}) + V_{cc}^2 \times f \times 10^{-6} [C_{pd} \times (\text{number of individual functions}) + C_L \times (\text{number of output pins})]$$

The default values for V_{cc} , C_L , C_{pd} , f , are as follows:

V_{cc} is the 5 V;

C_L is the 50 pF;

C_{pd} according to specification or catalogue;

f is the 50 MHz.

7.2.2.3.1.3 Bipolar, gallium arsenide, NMOS digital circuits

Calculate:

$$P = \frac{P_M(P_M + 3.5)}{3P_M + 5} \text{ (in Watts)}$$

where P_M = maximum continuous power = V_{cc} typical $\times$ I_{cc} maximum

7.2.2.3.1.4 Circuits with a standby mode (for example: MOS memories)

Calculate:

$$P = P_p \frac{d}{100} + P_r \times \left(1 - \frac{d}{100}\right)$$

in which

d is the activation ratio as a percentage;

P_p is the worst case power consumption (specified or published);

P_r is the power consumed in standby mode: $P_r = (V_{cc} \text{ typical}) \times (I_{cc} \text{ stand by})$.

7.3 The reliability model

7.3.1 General form of the model and definitions

- For an active device, the failure rate is noted λ and breaks down as :

$$\lambda = \lambda_{die} + \lambda_{package}$$

in which

$$\lambda_{die} = \lambda_{thermal \text{ effects}} + \lambda_{EOS \text{ effects}}$$

and

$$\lambda_{package} = \lambda_{thermomechanical \text{ effects}}$$

MATHEMATICAL MODEL :

$$\lambda = \left\{ \underbrace{\lambda_1 \times N \times e^{-0.35 \times a} + \lambda_2}_{\lambda_{die}} \times \left\{ \frac{\sum_{i=1}^y (\pi_i)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right\} + \underbrace{\left\{ 2.75 \times 10^{-3} \times \pi_\alpha \times \left(\sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68} \right) \times \lambda_3 \right\}}_{\lambda_{package}} + \underbrace{\left\{ \pi_I \times \lambda_{EOS} \right\}}_{\lambda_{overstress}} \right\} \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 λ_1 : per transistor base failure rate of the integrated circuit family. See Table 16.
 λ_2 : failure rate related to the technology mastering of the integrated circuit. See Table 16.
 N : number of transistors of the integrated circuit.
 a : [(year of manufacturing) – 1998].
 $(\pi_i)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the integrated circuit mission profile.
 τ_i : i^{th} working time ratio of the integrated circuit for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the integrated circuit. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the integrated circuit being in storage (or dormant). With $\tau_{on} + \tau_{off} = 1$
 π_α : influence factor related to the thermal expansion coefficients difference, between the mounting substrate and the package material.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
 λ_3 : base failure rate of the integrated circuit package. See Table 17a and 17b
 π_I : influence factor related to the use of the integrated circuit (interface or not).
 λ_{EOS} : failure rate related to the electrical overstress in the considered application..

Technological structure	Temperature factor π_i
MOS BiCMOS (low voltage)	$e^{\left[A \left(\frac{1}{328} - \frac{1}{273+t_j} \right) \right]}$ A=3480 ; (Ea=0.3 eV)
Bipolar BiCMOS (high voltage)	$e^{\left[A \left(\frac{1}{328} - \frac{1}{273+t_j} \right) \right]}$ A=4640 ; (Ea=0.4 eV)
AsGa Numerical	$e^{\left[A \left(\frac{1}{373} - \frac{1}{273+t_j} \right) \right]}$ A=3480 ; (Ea=0.3 eV)
AsGa MMIC	$e^{\left[A \left(\frac{1}{373} - \frac{1}{273+t_j} \right) \right]}$ A=4640 ; (Ea=0.4 eV)
t_j = Junction temperature in °C .	

Mathematical expression of the influence factor π_α	$\pi_\alpha = 0.06 \times (\alpha_S - \alpha_C)^{1.68}$
Mismatch between substrate and package for the thermal expansion coefficient	$ \alpha_S - \alpha_C $
α_S	See Table 14
α_C	See Table 14

Interface circuits Typical calculated values			λ_{EOS} FIT	π_I
Function	Electrical environment			
Interfaces	Computer		10	1
	Telecoms	switching	15	1
		transmitting, access, subscriber cards	40	1
		subscriber equipment	70	1
	Railways, payphone		100	1
	Civilian avionics (on board calculators)		20	1
	Voltage supply, Converters		40	1
Non Interfaces	All electrical environment		-	0

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ac})_i \right] - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.	

Table 14 – Thermal expansion coefficients α_s and α_c .

Linear thermal expansion coefficients	Material type	Values in ppm/°C
α_s (Substrate)	Epoxy Glass (FR4, G-10)	16
	PTFE Glass (polytetrafluoroethylene)	20
	Flexible substrate (Polyimide Aramid)	6.5
	Cu/Invar/Cu (20/60/20)	5.4
α_c (Component)	Epoxy (Plastic package)	21.5
	Alumina (ceramic package)	6.5
	Kovar (Metallic package)	5

Table 15 – Failure distribution (for non interfaces integrated circuits)

Environment type	Stuck at Valim in %	Stuck at ground %	Open circuit %
Ground; Benign	50	50	0
Ground; Fixed	5	5	90
Ground; mobile			

- For interface circuits, quasi totalities of defects are open circuits.

Calculation example:

A microprocessor failure rate has to be calculated for an "automotive passenger compartment" mission profile. This microprocessor has the following characteristics:

Manufacturing year : 1999 $a=1999-1998=1$ (See necessary information section)

Technology : numerical CMOS $\lambda_1=3.4 \times 10^{-6}$ and $\lambda_2=1.7$ (See Table 16)

Transistor number : 1.5×10^6

Dissipated power by the component : 0.5 W

PQFP 80 pins package $\alpha_c=21.5$ (See Table 14)

Mounting substrate FR4 $\alpha_s=16$ (See Table 14) Then $\pi_a=1$

This circuit is not an interface. Then $\pi_I=0$.

Temperature increase related to fact that the circuit works, is given with natural convection (then $K=1.4$) by the following formulas. Junction – ambient thermal resistance for the PQFP package is given in table 12 :

$$RTH_{ja} = (0.4 + 0.6 \times 1.4) \times \left(27 + \frac{2260}{80 + 3} \right) = 67^\circ C/W \quad \text{Then } \Delta T_j = 67 \times 0.5 = 34^\circ C$$

Mission profile from table 6 is following :

$$(t_j)_1 = 27 + 34 = 61^\circ C \quad \text{then} \quad (\pi_t)_1 = e^{\left[\frac{3480}{328} \left(\frac{1}{273+61} \right) \right]} = 1.21$$

$$(t_j)_2 = 30 + 34 = 64^\circ C \quad \text{then} \quad (\pi_t)_2 = e^{\left[\frac{3480}{328} \left(\frac{1}{273+64} \right) \right]} = 1.33$$

$$(t_j)_3 = 85 + 34 = 118^\circ C \quad \text{then} \quad (\pi_t)_3 = e^{\left[\frac{3480}{328} \left(\frac{1}{273+118} \right) \right]} = 5.5$$

$(t_{ae})_1$ is $14^\circ C$ for world wide climate

$(t_{ae})_1$ average temperature during the working phases is : $\frac{0.006 \times 27 + 0.046 \times 30 + 0.006 \times 85}{0.058} = 35^\circ C$

For the night starts phase, t_{ae} is $5^\circ C$; for the day light starts phase t_{ae} is $15^\circ C$, then :

$$\Delta T_1 = \left(\frac{34}{3} + 35 \right) - 5 = 41 \quad \Delta T_2 = \left(\frac{34}{3} + 35 \right) - 15 = 31 \quad \Delta T_3 = 10$$

$$\lambda = \left(\underbrace{\left[3.4 \times 10^{-6} \times 1.5 \times 10^6 \times e^{-0.35 \times 1} + 3.4 \right] \times \left[\frac{1.21 \times 0.006 + 1.33 \times 0.046 + 5.5 \times 0.006}{0.058 + 0.942} \right]}_{\lambda_{die}} + \left[\underbrace{2.75 \times 10^{-3} \times \left((670)^{0.76} \times (41)^{0.68} + (1340)^{0.76} \times (31)^{0.68} + (30)^{0.76} \times (10)^{0.68} \right)}_{\lambda_{package} \times 10.2} + \left[\frac{0 \times \lambda_{EQS}}{\lambda_{overstress}} \right] \right) \times 10^{-9} / h$$

The failure rate for this component will be 126 FIT for this mission profile.

Table 16 – Values of λ_1 and λ_2 for integrated circuits families

ABBREVIATIONS	TYPES	N Is the representative number of transistors	λ_1 in FIT	λ_2 in FIT
Silicon: MOS : Standard circuits (3)				
ROM DRAM/VideoRAM/AudioRAM High speed SRAM, FIFO Low consumption SRAM Double access SRAM EPROM, UVPROM, REPROM OTP FLASH EEPROM, flash EEPROM	Digital circuits, Micros, DSP	4 per gate	$3.4 \cdot 10^{-6}$	1.7
	Linear circuits	Actual number	$1.0 \cdot 10^{-2}$	4.2
	Digital / linear circuits (Telecom, CAN, CNA, RAMDAC, ...)	Actual number	$2.7 \cdot 10^{-4}$	20
	MEMORIES:			
	Read only memory	1 per bit	$1.7 \cdot 10^{-7}$	8.8
	Dynamic, Read Access Memory	1 per bit	$1.0 \cdot 10^{-7}$	5.6
	Static Read Access Memory - First in First out register; ("mixed MOS ")	4 per bit	$1.7 \cdot 10^{-7}$	8.8
	Static Read Access Memory - Low consumption; (CMOS)	6 per bit	$1.7 \cdot 10^{-7}$	8.8
	Double Access Static RAM	8 per bit	$1.7 \cdot 10^{-7}$	8.8
	Electrically programmable, UV erasable - Read only memory	1 /programmable point }	$2.6 \cdot 10^{-7}$	34
	One time programmable EPROM			
	Electrically programmable and erasable (block) (1)			
	Electrically programmable and erasable (word) (2)	2 /programmable point	$6.5 \cdot 10^{-7}$	16
(1) Whole memory array or blocks of words erasable (2) Blocks of words or word erasable (3) MOS include CMOS, HCMOS, NMOS, ... technologies				
Silicon: MOS : Asic circuits				
LCA (RAM based) PLD (GAL, PAL) (2) CPLD (EPLD, MAX, FLEX, FPGA, etc)	Standard Cell, Full Custom	4 per gate	$1.2 \cdot 10^{-3}$	10
	Gate Arrays	4 per gate	$2.0 \cdot 10^{-5}$	10
	USER PROGRAMMABLE LOGIC DEVICE:			
	Logic Cell Array electrically configured by external memory	40 per gate (1)	$4.0 \cdot 10^{-5}$	8.8
	Electrically Programmable and erasable (AND/OR array)	3 par grid point	$1.2 \cdot 10^{-3}$	16
	Electrically Programmable (interconnected macrocells array) (2)	100 per macrocell	$2.0 \cdot 10^{-5}$	34
(1) or 4000 per macrocell ; (2) EEPROM, EPROM, or Antifuse technologies.				
Silicon: Bipolar circuits (1)				
SRAM PROM, PLD (PAL)	Digital circuits	3 per gate	$6.0 \cdot 10^{-4}$	1.7
	Linear circuits (FET, others)	Actual number	$2.2 \cdot 10^{-2}$	3.3
	MMIC	Actual number	1.0	3.3
	Linear / Digital circuits, low voltage (<30V)	Actual number	$2.7 \cdot 10^{-3}$	20
	Linear / Digital circuits, high voltage(>=30V)	Actual number	$2.7 \cdot 10^{-2}$	20
	MEMORIES – PROGRAMMABLE ARRAYS- GATE ARRAYS:			
	Static read access memories	2.5 per bit	$3.0 \cdot 10^{-4}$	1.7
	Programmable read only memory	1.2 / , programmable point	$1.5 \cdot 10^{-4}$	32
	One time electrically programmable logic array (AND / OR arrays)	1.6 per grid point	$1.5 \cdot 10^{-4}$	32
	Gate arrays	3 per gate	$1.0 \cdot 10^{-3}$	10
Bipolar include : TTL, MTTL, LSTTL, FET, JFET, ECL, etc... technologies.				
Silicon: Bipolar and MOS circuits (BICMOS)				
SRAM	Digital circuits	4 per gate	$1.0 \cdot 10^{-6}$	1.7
	Linear / digital circuits low voltage (< 6V)	Actual number	$2.7 \cdot 10^{-4}$	20
	Linear / digital circuits, high voltage (>= 6V) and Smart Power	Actual number	$2.7 \cdot 10^{-3}$	20
	Static Read Access Memory	4 per bit	$6.8 \cdot 10^{-7}$	8.8
	Gate arrays	4 per gate	$6.4 \cdot 10^{-5}$	10
Gallium arsenide				
Digital	with only normally on transistors.	5 per gate	2.5	25
Digital	with normally off and normally on transistors.	3 per gate	$4.5 \cdot 10^{-4}$	16
MMIC	Low noise or low power (< 100mW) microwave circuits.	Actual number	2.0	20
MMIC	Power (> 100mW) microwave circuits.	Actual number	4.0	40

METHOD-1:

**Table 17a – λ_3 values for integrated circuits as a function of S
(pin number of the package)**

Abbreviation	Material type	Description		Pin number: <i>S</i>	λ_3 in FIT
SO ,SOP:1.27 mm pitch	Epoxy	Plastic Small Outline, L lead; Widths: 3.8 – 7.5 mm		4 to 40	$=0.012\times S^{1.65}$
Power SO	Epoxy	idem SO with heat sink			idem SO
SOJ: 1.27 mm pitch	Epoxy	Plastic Small Outline, J Lead; Width: 10.16 mm		28 to 44	$=0.023\times S^{1.5}$
VSOP: 0.76 mm pitch	Epoxy	Very Small Outline, L Lead; Width: 10.16 mm		40 to 56	$=0.011\times S^{1.47}$
SSOP: 0.65 mm pitch	Epoxy	Shrink Small Outline, L Lead; Width: 10.16 mm		8 to 56	$=0.013\times S^{1.35}$
TSSOP: 0.65 mm pitch	Epoxy	Thin Shrink Small Outline, L Lead; Widths: 4.1 - 6.1 mm		8 to 38	$=0.011\times S^{1.4}$
TSOP I: 0.55 mm pitch	Epoxy	Thin Small Outline, L Lead on small edge; Length: 11.8 mm		18 to 32	$=0.54\times S^{0.4}$
TSOP I: 0.5 mm pitch	Epoxy	Thin Small Outline, L Lead on small edge; Length: 18.4 mm		18 to 32	$=1.0\times S^{0.36}$
TSOP II: 0.8 mm pitch	Epoxy	Thin Small Outline, L Lead on long edge; Width: 10.16mm.		28 to 54	$=0.04\times S^{1.2}$
TSOP II: 0,65 mm pitch	Epoxy	Thin Small Outline, L Lead on long edge; Width: 10.16mm.		34 to 60	$=0.042\times S^{1.1}$
TSOP II: 0,5 mm pitch	Epoxy	Thin Small Outline, L Lead on long edge; Width: 10.16mm		34 to 60	$=0.075\times S^{0.9}$
TSOP II: 0,4 mm pitch	Epoxy	Thin Small Outline, L Lead on long edge; Width: 10.16mm		34 to 60	$=0.13\times S^{0.7}$
PLCC: 1,27 mm pitch	Epoxy	Plastic Leaded Chip Carrier, J Lead, all bodies		20 to 84	$=0.021\times S^{1.57}$
CLCC: 1,27 mm pitch	Alumina	Ceramic Leadless (and Leaded) Chip Carrier, all bodies			idem PLCC
MQUAD: 1,27 mm pitch	Kovar	Metallic Quad Flat Package (PLCC footprint); all bodies			idem PLCC
PQFP, TQFP	Epoxy	Plastic (Thin) Quad Flatpack, L Lead, Bodies defined in following column	5x5 mm ²	32 to 40	1.3
			10x10 mm ²	40 to 60	4.1
			14x14 mm ²	60 to 68	7.2
			14x20 mm ²	68 to 110	10.2
			28x28 mm ²	110 to 225	23
			32x32 mm ²	225 to 280	29
			40x40 mm ²	280 to 304	42
ED QUAD, Power QUAD	Epoxy	idem PQFP with heat sink (exposed slug)			idem PQFP
CQFP, CERQUAD	Alumina	Ceramic Quad Flat pack			idem PQFP
MQFP, MQUAD	Kovar	Metallic Quad Flat pack			idem PQFP
PBGA	Epoxy	Plastic Ball Grid Array- pas >1mm. Bodies defined in following column	13.5x15 mm ²	64 to 80	11.4
			17.4x19 mm ²	80 to 160	16.6
			23x23 mm ²	160 to 280	26.6
			35x35 mm ²	280 to 400	51.3
SBGA	Epoxy	Shrink BGA-pas 1mm-Corps 42.5x42.5 mm ²		580	71
SBGA	Epoxy	Shrink BGA-pas 1mm-Corps 27x27 mm ²		672	33
CBGA	Alumina	Ceramic			idem PBGA
PDIL	Epoxy	Palstic Dual In Line		8 to 64	$=9+0.09\times S$
CDIL, CERDIP	Alumina	Ceramic Dual In Line		8 to 64	$=9+0.09\times S$
PPGA	Epoxy	Plastic Pin Grid Array		40 to 160	$=9+0.09\times S$
CPGA	Alumina	Ceramic Pin Grid Array		40 to 160	$=9+0.09\times S$

METHOD-2:**Table 17b – λ_3 values for surface mounted integrated circuits packages as a function of D (package diagonal)**

Packages types	Examples	λ_3 in FIT
Two rows connections packages	SO; SOP; SOJ; VSOP; SSOP; TSSOP; TSOP I; TSOP II; etc...	$= 0.024 \times D^{1.68}$ (1)
Peripheral connections packages	PLCC; CLCC; MQUAD; PQFP; TQFP; CQFP; MQFP; etc...	$= 0.048 \times D^{1.68}$ (2)
Matrix connections packages	PBGA; CBGA; SBGA; μ BGA; CSP; etc...	$= 0.073 \times D^{1.68}$ (3)
Bare die with epoxy drop	COB (chip on board)	$= 0.048 \times D^{1.68}$ (4)
Note (1) : $D = \left[\left(\left(\frac{S}{2} - 1 \right) \times (pitch) \right)^2 + (Width)^2 \right]^{\frac{1}{2}}$ Note (2) : $D = \left[\left(\left(\frac{S}{4} - 1 \right) \times (pitch) \right)^2 + (Width)^2 \right]^{\frac{1}{2}}$ Note (3) : $D = \left[(Length)^2 + (Width)^2 \right]^{\frac{1}{2}}$ Note (4) : D report area diagonal		

8 Diodes and thyristors, transistors, optocouplers (IEC 60747-xx)

8.1 Evaluating the junction temperature of diodes and transistors

To evaluate the junction temperature of a transistor or diode, the preferred method for critical cases is to measure or calculate on the basis of thermal resistive network models. By default, the following simplified method will be used:

The junction temperature is given according to the average dissipated power, by one of the following equations:

$$\begin{aligned} t_j &= t_a + P \times R_{ja} \\ t_j &= t_c + P \times R_{jc} \end{aligned}$$

where:

t_a is the ambient temperature around the case (°C);

t_c is the case temperature (or socket temperature) (°C);

t_j is the junction temperature (°C);

R_{ja} is the Thermal resistance (junction-ambient) of the component (°C/W);

R_{jc} is the Thermal resistance junction-case or mounting base) (°C/W);

P is the average power dissipated (Watt).

Table 18 gives various thermal resistance values.

The first column gives the junction-case thermal resistance R_{jc} for power components or certain applications (such as use of a heat sink or insertion in a socket).

The second column gives the junction-ambient thermal resistance R_{ja} for components used in conditions of natural cooling, without additional heat sink.

The third column gives the junction-ambient thermal resistance for components soldered in a "surface mount" production line.

Thermal resistance depends largely on the type of case, but also on the size of the die and the internal fixing mode (bonding, hard solder, soft solder): the values given represent realistic examples.

Note the following equation:

$$\text{(junction - ambient) thermal resistance} = \text{(junction - case) thermal resistance} + \text{(case - ambient) thermal resistance}$$

**Table 18 – Values of λ_B and junction resistances
for active discrete components**

Package abbreviations	R _{jc} °C/W	R _{ja} °C/W	R _{ja} mounted component °C/W	λ_B FIT
TO-18	130	450		1
TO-39	35	200		2.0
TO-92	100	300		1
SOT-23			400	1.0
SOT-143			400	1.0
SOT-223			85	3.4
SOT-323			600	0.8
SOT-343			600	0.8
SOT-346			500	1
SOT-363			600	0.8
SOT-457			350	1.1
SOT-89			125	2.0
SOT-32 (TO-126)	10	100		5.3
SOT-82	10	100		5.3
DPACK (SOT428)			30	5.1
D2PACK			15	5.7
TO-220	3			5.7
TO-218 (SOT-93)	1.5			6.9
TO-247	1			6.9
ISOTOP	0.25			20.0
SOT-90B (optocoupler)		250		4.1
SO-8 (optocoupler)			300	4.5
DO-34 (DO-204AG)		500		2.5
DO-35 (DO-204AH)		400		2.5
DO-41 (DO-204AL) (glass)		150		2.5
DO-41 (DO-204AL) (plastic)		100		1
F 126		70		1
micromelf			600	2.5
SOD-80 (minimelf)			600	2.5
melf			450	5.0
SOD-110			350	0.8
SOD-123			600	1.0
SOD-323			600	0.7
SOD-523			100	0.5
SMA			600	1.8
SMB (DO-214)			75	2.4
SMC (DO-215)			25	5.1
DO-220	3			5.7
SOD-15			20	5.1

8.2 Low power diodes

Silicon: signal diodes; PIN; fast and slow recovery rectifier diodes; SCHOTTKY; up to 3A;
Thyristors, triacs up to 3A;
Zener diodes up to 1,5W;
Transient voltage suppressors, up to 5 kW (peak, 10µs/1000µs);
Gallium arsenide diodes, up to 0.1W.

MATHEMATICAL MODEL

$$\lambda = \left\{ \underbrace{\pi_U \times \lambda_0}_{\lambda_{die}} \times \left\{ \frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right\} + \left\{ \underbrace{2.75 \times 10^{-3} \times \sum_{i=1}^y (\pi_n)_i \times (\Delta T_i)^{0.68}}_{\lambda_{package}} \times \lambda_B \right\} + \left\{ \frac{\pi_I \times \lambda_{EOS}}{\lambda_{overstress}} \right\} \right\} \times 10^{-9} / h$$

NECESSARY INFORMATION:

$(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

π_U : use factor (permanent or not);

λ_0 : base failure rate of the die. See table on this page;

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the diode mission profile;

τ_i : i^{th} working time ratio of the diode for the i^{th} junction temperature of the mission profile;

τ_{on} : total working time ratio of the diode. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

τ_{off} : time ratio for the diode being in storage (or dormant) mode;

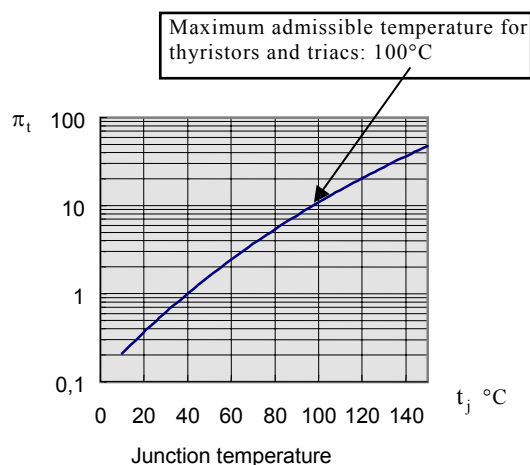
$(\pi_n)_i$: i^{th} influence factor related to the annual number cycles of thermal variations seen by the package, with the amplitude ΔT_i

ΔT_i : i^{th} thermal amplitude variation of the mission profile;

λ_B : base failure rate of the diode package. See Table 18;

π_I : influence factor related to the use of the diode (protection interface or not);

λ_{EOS} : failure rate related to the electrical overstress in the considered application.



For protection diodes tj = ambient temperature
other diodes tj = ambient temp. +(R_{th} × P)

Mathematical formula for π_t

$$\pi_t = e^{4640 \left(\frac{1}{313} - \frac{1}{t_j + 273} \right)} \quad (0.4 \text{ electron-volt})$$

Protection diodes as interface; typical calculated values			λ _{EOS} FIT	π _I
Function	Electrical environment			
Protection Interface	Computer		10	1
	Telecoms	switching	15	1
		transmitting access, subscriber cards	40	1
		Subscriber equipment	70	1
	Railways, payphone		100	1
	Civilian avionics (on board calculators)		20	1
	Voltage supply, Converters		40	1
Non Interfaces	All electrical environment			0

Failure rate for $t_j=40^{\circ}\text{C}$ (expressed in FIT)			λ_0
Silicon diodes	signal ($<1\text{A}$)		0.07
	Recovery , rectifier 1A à 3A		0.1
	Zener (regulator) ≤ 1.5 watt		0.4
	suppressor	Transient voltage	2.3
		Trigger transient voltage	2
Gallium arsenide diodes ($\leq 0.1\text{ W}$)			0.3
Thyristors, triacs ($\leq 3\text{A}$)			1

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ac})_i \right] - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.	

Type of use		π_U
Thyristors and triacs	Permanent use*	10
	Occasional use* (ratio off/on > 1)	1
Other diodes		1
*: The expression "use" corresponds to a switching condition with component reverse biased part of the time . When the component is forward biased permanently, take $\pi_U = 1$.		

Failure distribution			
	Zener diodes	Thyristors	Others
Short circuits	70%	20%	80%
Open circuits	20%	20%	20%
Zener voltage drift	10%		
Forward leakage current drift		60%	

NOTE 1 Predicted values can be reached by Schottky diodes only if it is a Platinum-Nickel structure.

NOTE 2 Diodes used as a protection function must be specified for this function (voltage suppressor). In the contrary case, the given failure rate λ_{EOS} for the electrical environment might be higher.

8.3 Power diodes

Silicon: Rectifier diodes; fast recovery rectifier diodes; SCHOTTKY; above 3A.

Thyristors, triacs; above 3A.

Zener diodes above 1,5 W.

Transient voltage suppressors 5 kW (en crête, 10µs/1000µs).

Gallium arsenide diodes, above 0,1W.

} excepted modules

MATHEMATICAL MODEL

$$\lambda = \left\{ \underbrace{\pi_U \times \lambda_0}_{\lambda_{die}} \times \left\{ \frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right\} + \left\{ \underbrace{2.75 \times 10^{-3} \times \sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68}}_{\lambda_{package}} \times \lambda_B \right\} + \left\{ \frac{\pi_I \times \lambda_{EOS}}{\lambda_{overstress}} \right\} \right\} \times 10^{-9} / h$$

NECESSARY INFORMATION:*

$(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

π_U : use factor (permanent or not).

λ_0 : base failure rate of the die. See table on this page.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the diode mission profile.

τ_i : i^{th} working time ratio of the diode for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the diode. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

τ_{off} : time ratio for the diode being in storage (or dormant) mode.

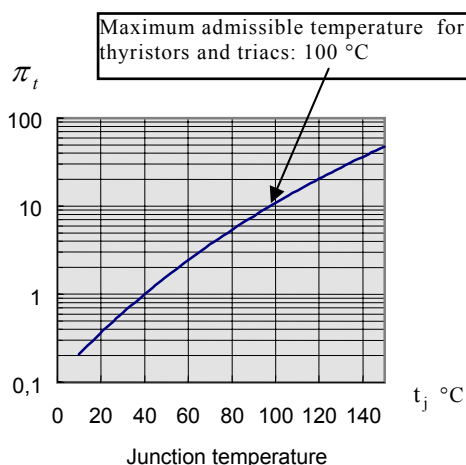
$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

λ_B : base failure rate of the diode package. see Table 18.

π_I : influence factor related to the use of the diode (protection interface or not).

λ_{EOS} : failure rate related to the electrical overstress in the considered application..



For protection diodes t_j = ambient temperature
other diodes t_j = ambient temp.
+ $(R_{th} \times P)$

Mathematical formula for π_t

$$\pi_t = e^{4640 \left(\frac{1}{313} - \frac{1}{t_j + 273} \right)} \quad (0.4 \text{ electron-volt})$$

Protection diodes as interface; typical calculated values			λ_{EOS} FIT	π_I
Function	Electrical environment			
Protection Interface	Computer		10	1
	Telecoms	switching	15	1
		transmitting access, subscriber cards	40	1
		Subscriber equipment	70	1
	Railways, payphone		100	1
	Civilian avionics (on board calculators)		20	1
Non Interfaces	Voltage supply, converters		40	1
	All electrical environment			0

Failure rate for $t_j=40^{\circ}\text{C}$ (expressed in FIT)			λ_0
Silicon Diodes	Recovery , rectifier (>3A)		0.7
	Zener (regulator) >1,5 watt		0.7
	suppressor	Transient voltage	0.7
		Trigger transient voltage	3
Gallium arsenide diodes (>0,1 w)			1
Thyristors, triacs (>3A)			3

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ac})_i \right] - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.	

Type of use		π_U
Thyristors and triacs	Permanent use*	10
	Occasional use* (ratio off/on>1)	1
Other diodes		1
*: The expression " use " corresponds to a switching condition. The component is reverse biased part of the time . When the component is forward biased permanently, take $\Pi_U = 1$.		

Failure distribution			
	Zener diodes	Thyristors	Others
Short-circuits	70%	20%	80%
Open-circuits	20%	20%	20%
Zener voltage drift	10%		
Forward leakage current drift		60%	

NOTE1 Predicted values can be reached by Schottky diodes only if it is a Platinum-Nickel structure.

NOTE 2 Diodes used as a protection function must be specified for this function (voltage suppressor). In the contrary case, the given failure rate λ_{EOS} for the electrical environment might be higher.

8.4 Low power transistors

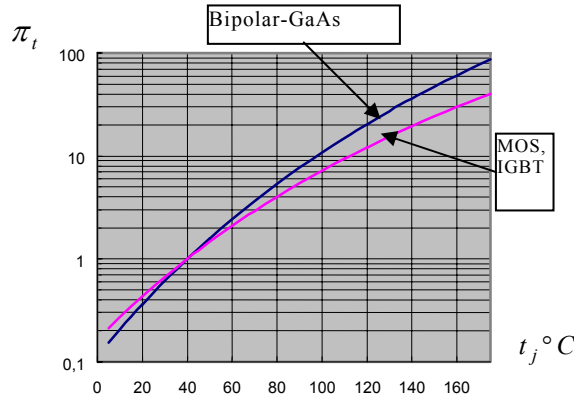
Silicon, junction, FET, MOS; up to 5 W.
Gallium Arsenide ; up to 1 W.

MATHEMATICAL MODEL

$$\lambda = \left\{ \underbrace{\pi_S \times \lambda_0}_{\lambda_{die}} \times \underbrace{\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}}}_{\lambda_{package}} + \underbrace{2.75 \times 10^{-3} \times \left(\sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68} \right)}_{\lambda_{package}} \times \lambda_B + \underbrace{\pi_I \times \lambda_{EOS}}_{\lambda_{overstress}} \right\} \times 10^{-9} / h$$

NECESSARY INFORMATION:

- (t_{ae})_i : average outside ambient temperature surrounding the equipment, during the ith phase of the mission profile.
 (t_{ac})_i : average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 π_S : charge factor.
 λ_0 : base failure rate of the die. See table on this page.
 (π_t)_i : ith temperature factor related to the ith junction temperature of the transistor mission profile.
 τ_i : i^{eme} working time ratio of the transistor for the ith junction temperature of the mission profile.
 τ_{on} : total working time ratio of the transistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the transistor being in storage (or dormant) mode.
 (π_n)_i : ith influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : ith thermal amplitude variation of the mission profile.
 λ_B : base failure rate of the transistor package. See Table 18.
 π_I : influence factor related to the use of the transistor (protection interface or not).
 λ_{EOS} : failure rate related to the electrical overstress in the considered application..

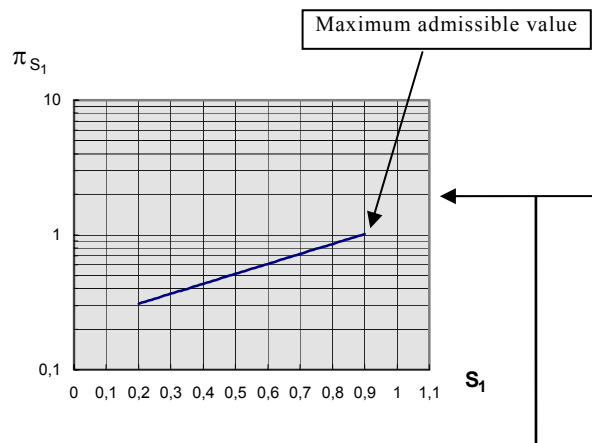


Protection transistor as interface; typical calculated values			λ_{EOS}	π_I
Function	Electrical environment		FIT	
Protection Interface	Computer		10	1
	Telecoms	switching	15	1
		transmitting access, subscriber cards	40	1
		Subscriber equipment	70	1
	Railways, payphone		100	1
	Civilian avionics (on board calculators)		20	1
	Voltage supply, converters		40	1
Non Interfaces	All electrical environment			0

Mathematical formulas for π_t and π_S		
π_t	Bipolar GaAs	$\pi_t = e^{\frac{4640}{373} \left(\frac{1}{t_j} - \frac{1}{273} \right)}$ (activation energy: 0.4 eV)
	MOS IGBT	$\pi_t = e^{\frac{3480}{373} \left(\frac{1}{t_j} - \frac{1}{273} \right)}$ (activation energy: 0.3 eV)
π_S	FET, MOS IGBT	$\pi_{S1} = 0.22e^{1.7S_1}$ $\pi_{S2} = 0.22e^{3S_2}$
	Bipolar	$\pi_S = 0.22e^{1.7S}$

Failure rate for $t_j=40^\circ\text{C}$ (expressed in FIT)	λ_0
Silicon transistors • Bipolar; npn; pnp • MOS p, n ; FET	0,75
Gallium Arsenide transistors	0,3

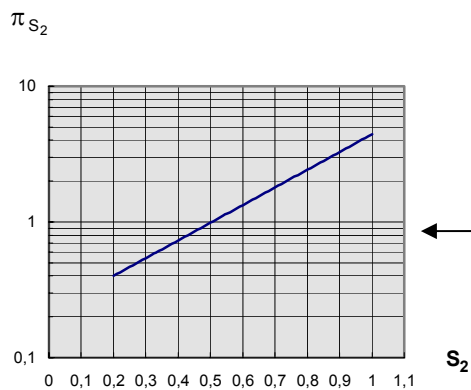
Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ac})_i \right] - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i th phase of the mission profile	



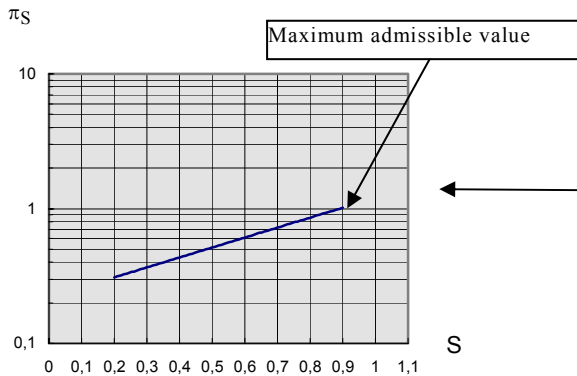
FET, MOS:
 $\pi_S = \pi_{S_1} \times \pi_{S_2}$

$$S_1 = \frac{\text{Maximum repetitive applied } V_{DS} \text{ voltage}}{\text{Minimum specified } V_{DS} \text{ voltage}}$$

- **S₂** is accepted up to 1 since the destructive gate oxide voltage value is normally three times greater than the maximum rated V_{gs} voltage (for MOS transistors).
- These values (maximum and destructive V_{gs} values) shall be checked during the qualification process. For instance, if the maximum rated V_{gs} voltage is 20 V, the destructive V_{gs} voltage value must be greater than 60 V (practically between 60 V and 80 V).



$$S_2 = \frac{\text{Maximum repetitive applied } V_{GS} \text{ voltage}}{\text{Minimum specified } V_{GS} \text{ voltage}}$$



BIPOLAR

$$S = \frac{\text{Maximum repetitive applied } V_{CE} \text{ voltage}}{\text{Minimum specified } V_{CE} \text{ breakdown voltage}}$$

Fragile assembly:
1 – Cavity package weighting more than 0,1 g
2 – Package weighting more than 0,1g and fixed by leads (more than 2 mm)
Strong assembly:
1 – Molded plastic components (lead less than 2 mm) or DIL, or surface mounted component (SMT)
2 – Firmly fixed packages (screw, clips, glue)
3 – Components weighting more than 0,1 g

Failure distribution		%
Silicon	Short-circuits	85
	Open-circuits	15
Gallium - arsenide	Short-circuits	95
	Open-circuits	5

8.5 Power transistors

Silicon: 5 W or more

Gallium arsenide: 1 W or more

{ Excepted modules

CAUTION !

Under cyclic operation, the life time is limited !

MATHEMATICAL MODEL

$$\lambda = \left\{ \underbrace{\pi_S \times \lambda_0}_{\lambda_{die}} \times \underbrace{\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}}}_{\lambda_{package}} + \left\{ \underbrace{2.75 \times 10^{-3} \times \sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68}}_{\lambda_B} \times \lambda_B \right\} + \underbrace{\left\{ \frac{\pi_I \times \lambda_{EOS}}{\lambda_{overstress}} \right\}}_{\lambda_{overstress}} \right\} \times 10^{-9} / h$$

NECESSARY INFORMATION:

$(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

π_S : charge factor.

λ_0 : base failure rate of the die. See table on this page.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the transistor mission profile.

τ_i : i^{th} working time ratio of the transistor for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the transistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

τ_{off} : time ratio for the transistor being in storage (or dormant) mode.

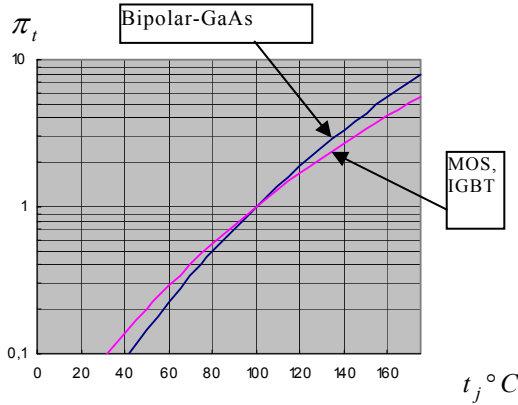
$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

λ_B : base failure rate of the transistor package. See Table 18.

π_I : influence factor related to the use of the transistor (protection interface or not).

λ_{EOS} : failure rate related to the electrical overstress in the considered application.

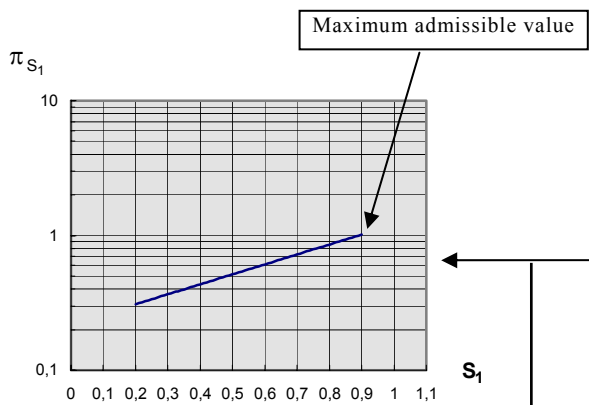


Mathematical formulas for π_t and π_S		
π_t	Bipolar GaAs	$\pi_t = e^{4640(\frac{1}{373} - \frac{1}{t_j + 273})}$ (activation energy : 0.4 ev)
	MOS IGBT	$\pi_t = e^{3480(\frac{1}{373} - \frac{1}{t_j + 273})}$ (activation energy : 0.3 ev)
π_S	FET, MOS IGBT	$\pi_{S_1} = 0.22e^{1.7S_1}$ $\pi_{S_2} = 0.22e^{3S_2}$
	Bipolar	$\pi_S = 0.22e^{1.7S}$

Protection transistor as interface; typical calculated values			λ_{EOS}	π_I
Function	Electrical environment		FIT	
Protection Interface	Computer		10	1
	Telecoms	switching	15	1
		transmitting access, subscriber cards	40	1
		Subscriber equipment	70	1
	Railways, payphone		100	1
	Civilian avionics (on board calculators)		20	1
	Voltage supply, converters		40	1
Non Interfaces	All electrical environment			0

Failure rate for $t_j=100\text{ }^{\circ}\text{C}$ (expressed in FIT)	λ_0
Silicon transistors • Bipolar; npn; pnp • MOS p, n ; FET	2
Gallium Arsenide transistors	1

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase		$\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ac})_i \right] - (t_{ac})_i$
For a permanent working phase, storage or dormant		ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile



Life expectancy
Under cyclic operation, the number of cycles is limited to:

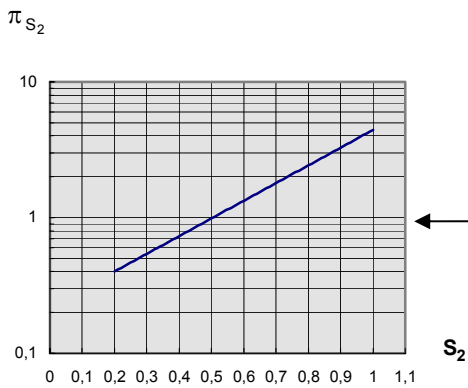
$$N = 10^7 e^{-0.05 \Delta t_j}$$

Δt_j : junction temperature range

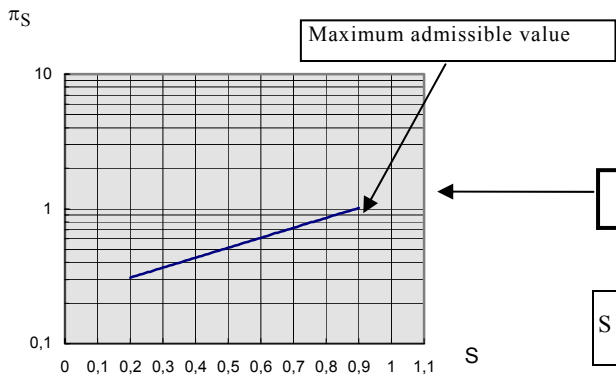
FET, MOS:
 $\pi_S = \pi_{S_1} \times \pi_{S_2}$

$$S_1 = \frac{\text{Maximum repetitive applied } V_{DS} \text{ voltage}}{\text{Minimum specified } V_{DS} \text{ voltage}}$$

- **S₂** is accepted up to 1 since the destructive gate oxide voltage value is normally three times greater than the maximum rated V_{gs} voltage (for MOS transistors).
- These values (maximum and destructive V_{gs} values) shall be checked during the qualification process. For instance, if the maximum rated V_{gs} voltage is 20 V, the destructive V_{gs} voltage value must be greater than 60 V (practically between 60 V and 80 V).



$$S_2 = \frac{\text{Maximum repetitive applied } V_{GS} \text{ voltage}}{\text{Minimum specified } V_{GS} \text{ voltage}}$$



BIPOLAR

$$S = \frac{\text{Maximum repetitive applied } V_{CE} \text{ voltage}}{\text{Minimum specified } V_{CE} \text{ breakdown voltage}}$$

- Fragile assembly:**
- 1 – Cavity package weighting more than 0,1 g
 - 2 – Package weighting more than 0,1 g and fixed by leads (more than 2 mm)
- Strong assembly:**
- 1 – Molded plastic components (lead less than 2 mm) or DIL, or surface mounted component (SMT)
 - 2 – Firmly fixed packages (screw, clips, glue)
 - 3 – Components weighting more than 0,1 g

Failure distribution		%
Silicon	Short-circuits	85
	Open-circuits	15
Gallium - arsenide	Short-circuits	95
	Open-circuits	5

8.6 Optocouplers

8.6.1 Life expectancy

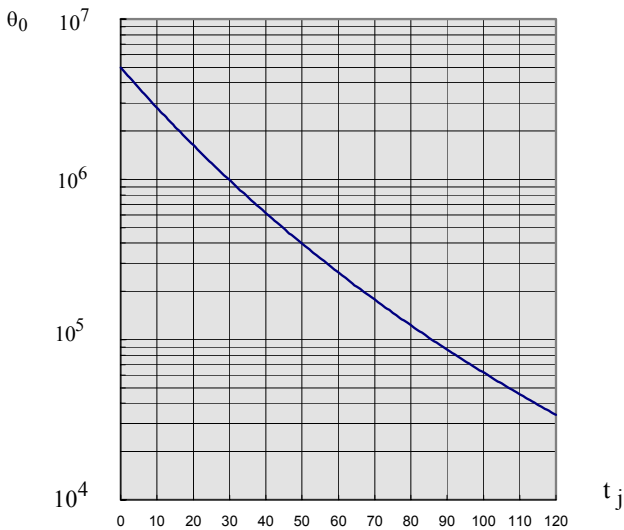
The operating time must not exceed the life expectancy value θ . Beyond this time, the failure rate λ cannot be assumed to be constant.

$$\theta = \theta_0 \times \kappa_0 \times \kappa_1 \times \kappa_2 \times \kappa_3 \quad \text{h}$$

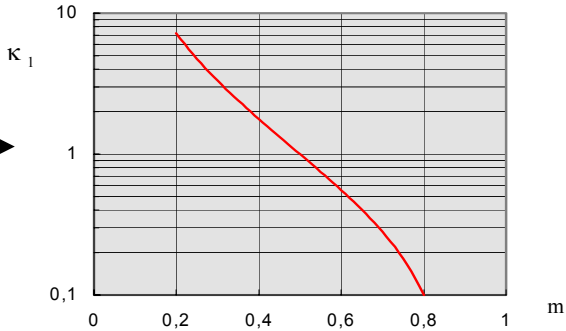
For molded plastic optocouplers, the temperature junction must not exceed 90 °C

Necessary information	For
Junction temperature t_j	θ_0
Input current (operating and test condition)	κ_0, κ_2
Initial, final transfer ratio	κ_1

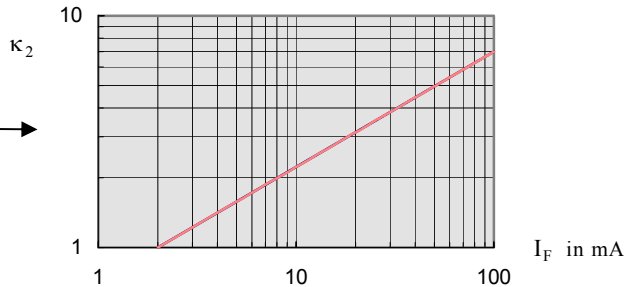
$$\kappa_0 = \frac{50}{\text{input current (operating, mA)}}$$



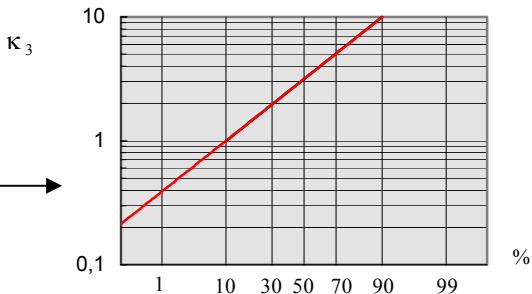
Life expectancy in hours= f(junction temperature t_j)



$$\kappa_1 \text{ according to the selected ratio } m = \frac{\text{Final transfert ratio}}{\text{Initial trans fer ratio}}$$



$$\kappa_2 \text{ according to the measured current } I_F \text{ of the transfer ratio (testing conditions).}$$



$$\kappa_3 \text{ for a cumulative failure ratio different from 10 \% .}$$

Optocoupler life expectancy θ_0 according to the junction temperature and the following failure criterion:

- Transfer ratio measurement conditions
 $V_{CE} = 5 \text{ volts } I_F = 2\text{mA}$
 $\frac{\text{Final transfert ratio}}{\text{Initial transfert ratio}} = m = 0.5$
- For an operating current of 50 mA

Note The life expectancy is estimated for a cumulative failure ratio of 10 %.

Life expectancy θ_0

$$0.4e^{\frac{4640}{t_j+273}}$$

8.6.2 Failure rate

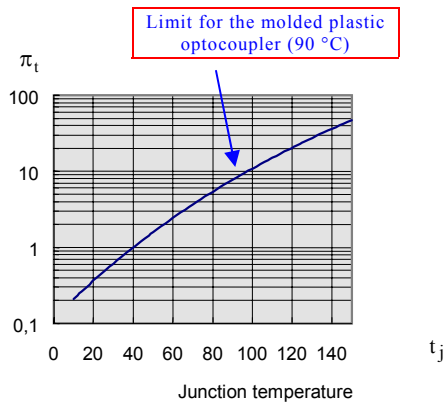
MATHEMATICAL MODEL

$$\lambda = \left\{ \underbrace{2.2 \times \pi_S}_{\lambda_{die}} \times \underbrace{\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}}}_{\lambda_{package}} + \left\{ \underbrace{2.75 \times 10^{-3} \times \sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68}}_{\lambda_{package}} \times \lambda_B \right\} + \underbrace{\left\{ \pi_I \times \lambda_{EOS} \right\}}_{\lambda_{overstress}} \right\} \times 10^{-9} / h$$

CAUTION !
Life expectancy is limited !

NECESSARY INFORMATION:

- $(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 π_S : charge factor.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the optocoupler mission profile.
 τ_i : i^{th} working time ratio of the optocoupler for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the optocoupler. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the optocoupler being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
 λ_B : base failure rate of the optocoupler package. See Table 18.
 π_I : influence factor related to the use of the optocoupler (interface or not).
 λ_{EOS} : failure rate related to the electrical overstress in the considered application.



$$t_j = \text{ambient temperature} + (\text{power dissipation}) \times R_{ja}$$

* emitter + receiver

$$\frac{V_{io}}{V_{ioRM}} \quad \begin{array}{l} \text{permanent applied voltage (input/output)*} \\ \text{peak, repetitive insulation voltage} \end{array}$$

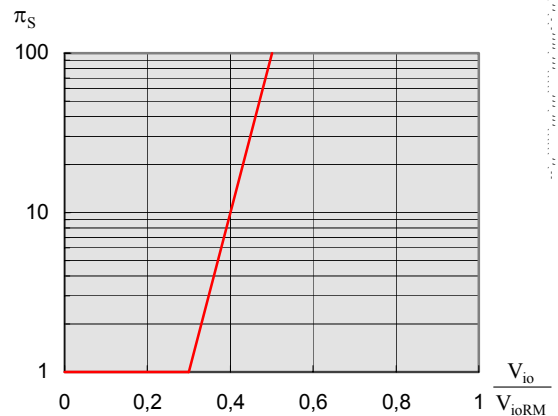
* without considering partial discharges.

Failure distribution	
Open circuits	50 %
Short circuits	10 %
Drift	40 %

Mathematical formula π_t

$$\pi_t = e^{4640 \left(\frac{1}{313} - \frac{1}{t_j + 273} \right)} \quad (\text{activation energy : 0.4 eV})$$

Optocoupler as interface; typical calculated values			λ_{EOS}	π_I
Function	Electrical environment		FIT	
Protection Interface	Computer		10	1
	Telecoms	switching	15	1
		transmitting access, subscriber cards	40	1
		Subscriber equipment	70	1
	Railways, payphone		100	1
	Civilian avionics (on board calculators)		20	1
	Voltage supply, converters		40	1
Non Interfaces	All electrical environment			0



Influence of the applied voltage between input-output

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[\frac{\Delta T_j}{3} + (t_{ae})_i \right] - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i =average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.	

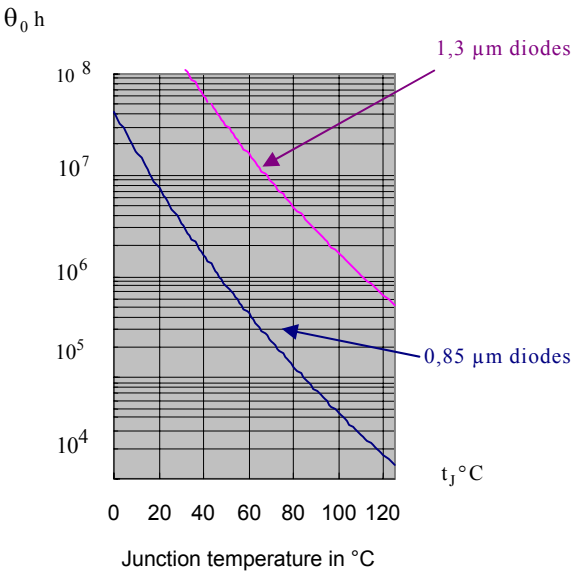
9 Optoelectronics

NOTE Optical component failure rates in this clause are only given for the ground fixed environment, with a permanent working mission profile, and a daylight-night thermal cycling of less than 3 °C.

9.1 Light emitting diodes diode modules (IEC 60747-12-2, IEC 62007)

9.1.1 Life expectancy

$$\theta = \theta_0 \times \kappa_0 \times \kappa_1 \times \kappa_2 \times \kappa_3 \quad \text{h}$$



Life expectancy θ_0 of a LED according to the junction temperature t_j

- For an operating current of 100 mA
- For the failure criterion:
$$m = \frac{\text{Final optical power}}{\text{Initial optical power}} = 0,5$$
(with an optical power measuring current equal to 100 mA)

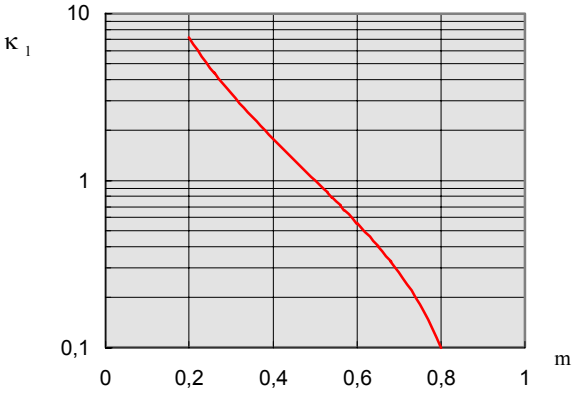
NOTE The life expectancy θ_0 is evaluated for a cumulative failure ratio of 10 %.

Diode types	Life expectancy θ_0 expressed in hours
0,85 μm diodes	$2,3 \times 10^5 e^{7000 \left(\frac{1}{t_j + 273} - \frac{1}{343} \right)} (0,6 \text{ eV})$
1,3 μm diodes	$8,7 \times 10^6 e^{7000 \left(\frac{1}{t_j + 273} - \frac{1}{343} \right)} (0,6 \text{ eV})$

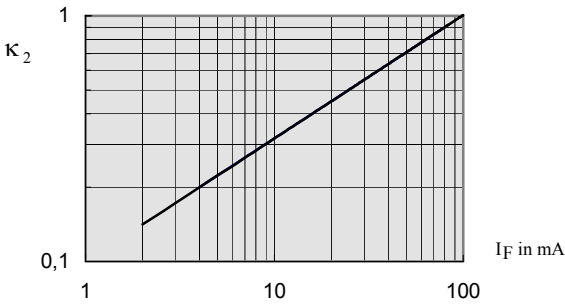
The operating time must not exceed the life expectancy value θ . Beyond this time the failure rate λ cannot be assumed to be constant.

Necessary information	For
Junction temperature t_j	θ_0
Operating current	κ_0
Measurement current (for optical power)	κ_2

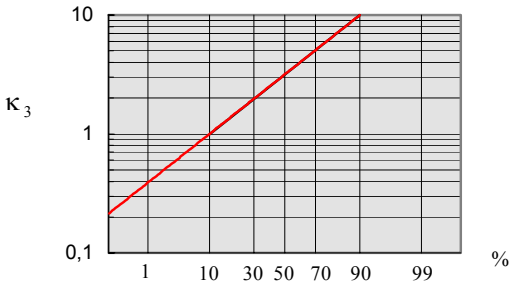
$$\kappa_0 = \frac{100}{\text{operating input current(mA)}}$$



κ_1 according to the selected ratio m:
with $m = \frac{\text{Final optical power}}{\text{Initial optical power}}$



κ_2 according to the optical power measuring current



κ_3 for a cumulative failures different from 10 %

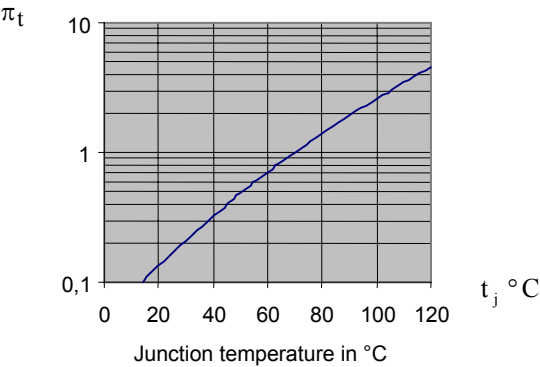
9.1.2 Failure rate (Modules datacom)

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \pi_t \times 10^{-9} \quad /h$$

CAUTION !

The life expectancy is limited !



Necessary information		for
Junction temperature	t _j	π _t
Module type		λ ₀

Module type	λ ₀ FIT
Elementary emitter module with fibered DEL without electronic	100
Emitter module with fibered DEL and DEL driver	130
Emitter / receiver module with fibered DEL + PIN + electronics with or without clock recovery.	180
Emitter / receiver module with fibered DEL +APD + electronics with or without clock recovery	200

Temperature influence

$$t_j = t_c + R_{th} \times P$$

P = applied power
t_c = case temperature
R_{th} = thermal resistance (junction/case)

- measured
- or rated
- by default, 150°C/W

Mathematical formula for π_t

$$\pi_t = e^{4060 \left(\frac{1}{343} - \frac{1}{t_j + 273} \right)}$$

(activation energy: 0.35eV)

Failure distribution according to package type	With window	With fibre
Short circuit (forward degradation)	70%	40%
Open	10%	10%
Optical coupling, or fiber	20%	50%

$\theta = \theta_0 \cdot \kappa_1 \cdot \kappa_2$ in hours

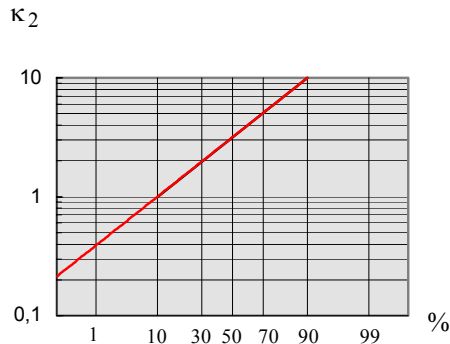
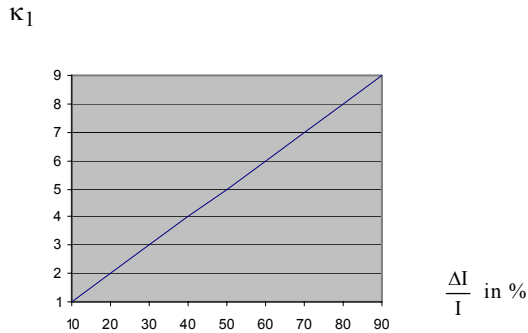
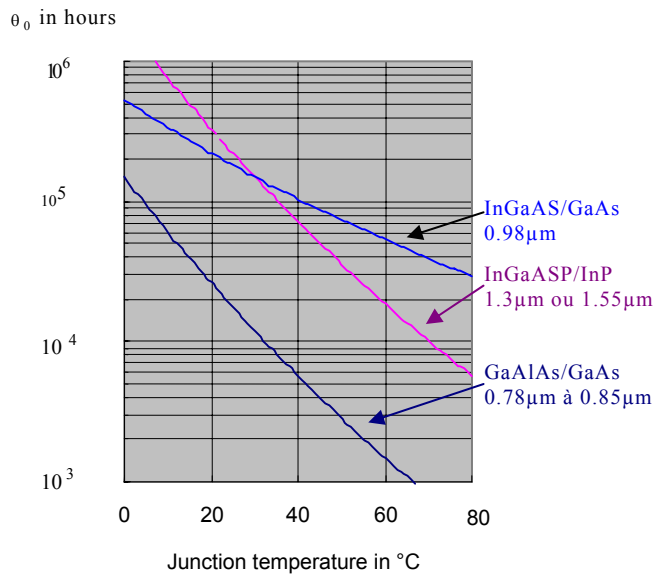
κ_1 is function of the failure criterion, which is the shift of the operating current at constant optical power, $\frac{\Delta I}{I}$ in %

- For emitter laser of numerical systems and pump lasers, take a criterion of 50%
- For emitter laser of analog systems and WDM, take 10%

κ_2 is function of the cumulative failures at the life expectancy value.

- Take generally 10%

The failure rate λ is assumed to be constant (after the infant mortality time), but only within the life expectancy limit θ .



Diode Types	Life expectancy θ_0
GaAlAs/GaAs diode (0.78m ; 0.85 μ m)	$4 \times 10^3 e^{7000 \left(\frac{1}{t_j+273} - \frac{1}{318} \right)}$ (0.6 eV)
InGaAsP/InP diodes (1.2 μ m ; 1.6 μ m)	$1.5 \times 10^5 e^{7000 \left(\frac{1}{t_j+273} - \frac{1}{303} \right)}$ (0.6 eV)
InGaAs/GaAs diodes (0.98 μ m)	$1.5 \times 10^5 e^{3500 \left(\frac{1}{t_j+273} - \frac{1}{303} \right)}$ (0.3 eV)

Note: LASER diodes are sensitive to electrostatic discharges (ESD), consequently their reliability is reduced, even with a low discharge.

9.2 Laser diodes modules - Failure rate

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \pi_t \times 10^{-9} \quad /h$$

CAUTION !

The life expectancy is limited !

GaAlAs/ GaAs

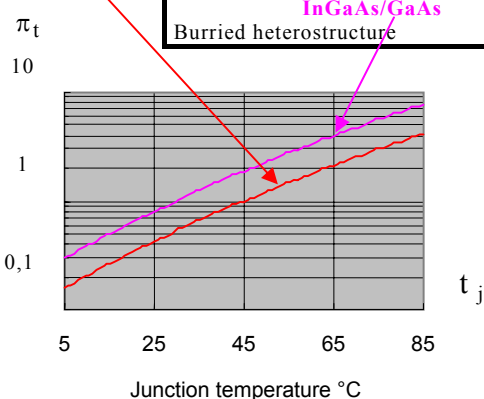
- gain guiding (transmitting ; 0,84 µm)
- index guiding (compact disks ; 0,8 µm ; with window)

InGaAs/InP

Burried heterostructure ; P  rot-Fabry or distributed feedback (DFB) 1,3 µm or 1,55 µm

InGaAs/GaAs

Burried heterostructure



Temperature influence

$$t_j = (\text{submount or case temperature}) + R_{th} \times (P_e - P_0)$$

R_{th} = thermal resistance (junction/case or junction/submount)

P_e = applied electrical power (W)

P_0 = optical power (W) (two facets)

Mathematical formulas for π_t

GaAlAs/GaAs	$\pi_t = e^{4060 \left(\frac{1}{318} - \frac{1}{t_j + 273} \right)}$
InGaAsP/InP	$\pi_t = e^{4060 \left(\frac{1}{303} - \frac{1}{t_j + 273} \right)}$
InGaAs/GaAs	$\pi_t = e^{4060 \left(\frac{1}{303} - \frac{1}{t_j + 273} \right)}$

Necessary information	For
Junction temperature	t_j
Type of material, internal structure.	$\pi_t ; \lambda_0$
Type of module	λ_0

Material	Window µm	Module type	λ_0 FIT
GaAlAs/GaAs	0,8	Elementary emitter modules *	3 000
InGaAs/InP	1,2 to 1,6	Elementary emitter modules * without electronics	40
InGaAs/InP	1,2 to 1,6	Emitter module with electronics.	60
InGaAs/InP	1,2 to 1,6	Emitter/receiver module, with laser PIN and electronics, with or without clock recovery (without crystal)	80
InGaAs/InP	1,2 to 1,6	Integrated modulator laser module	100
InGaAs/InP	1,48	Pump laser module Power ≤ 250 mW	200
InGaAs/InP	1,48	Pump laser module Power > 250 mW.	350
InGaAs/GaAs	0,98	Pump laser module	300

- * Generally an elementary laser module is made of a control photodiode, a laser diode and a coupling element. For other more complex structures, the failure rates mentioned in this table, include all the other elements excepted the thermoelectric cooler and the thermistor.

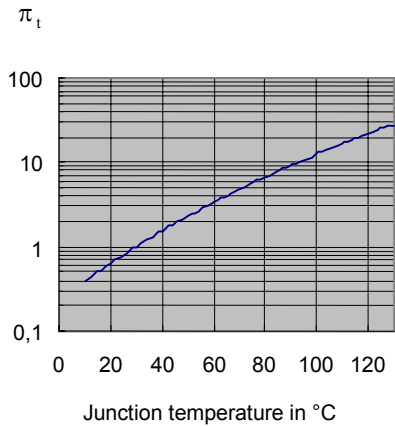
Failure modes according to the module type

Module type	Failure mode	Description	Failure rate	
			Modules transmission	Compact disks
1,3 µm/1,55 µm modules (monomode fibre 9 / 125)	Diodes failure	• Degradation of the spectrum • Current increase	10 %	
	Coupling failure	• High drop in output power	90 %	
Pump laser modules (0,98 µm 1,48 µm)	Diodes failure	• High current increase	90 %	
	Coupling failure	• High drop in output power	10 %	
0,85 µm modules (monomode fibre 9 / 125)	Diodes failure	• No laser effect • Degradation of the spectrum • Current increase	80 %	100 %
	Coupling failure	• High drop in output power	10 %	0 %
	Broken fibre	• No output power	10 %	0 %

9.3 Photodiodes and receiver modules for telecommunications (IEC 60747-12)

MATHEMATICAL MODEL
$\lambda = \lambda_0 \times \pi_t \times 10^{-9} \quad /h$

Necessary information	For
Junction temperature t_j	π_t
Type (silicon, GaAlAs, PIN , APD, module)	λ_0



Junction temperature t_j is admitted to be equal to the ambient temperature

Temperature influence

Type		λ_0 FIT
PIN diodes	Silicon (0,7 μm -1,1 μm window)	5
	InGaAs (1,2 μm -1,6 μm window)	10
APD diodes (avalanche)	Silicon	20
	Germanium	40
	InGaAs	80
PIN module + electronics with or without clock recovery		30
APD module + electronics with or without clock recovery		100

Mathematical formula for π_t
$\pi_t = e^{4060 \left(\frac{1}{303} - \frac{1}{t_j + 273} \right)}$
0.35 electron – volt

Failure modes according to package types	With window	With fibre
Short-circuit (reverse degradation)	80 %	40 %
Open circuit	20 %	10 %
Coupling	0 %	50 %

*Values are given for a starting production.

10 Capacitors and thermistors (NTC)

10.1 Fixed plastic, paper, dielectric capacitors - Radio interference suppression capacitors (plastic, paper)

MATHEMATICAL MODEL

$$\lambda = 0.1 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / \text{h}$$

NECESSARY INFORMATION:

$(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.

τ_i : i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.

$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

Continuous voltage applied to the capacitor

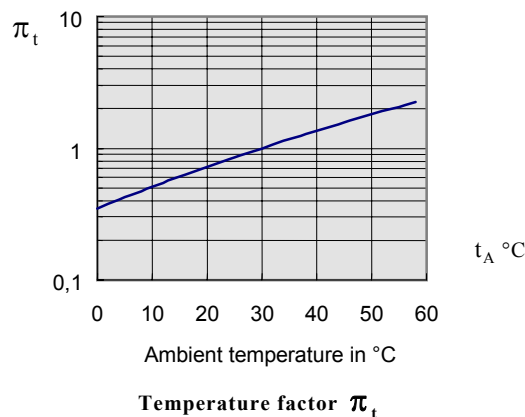
See note*

Peak value of the alternative voltage applied to the capacitor

See note*

Rated voltage of the capacitor.

See note*



Failure modes

Short-circuits	10 %
Open-circuits	90 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase		$\Delta T_i = (t_{ac})_i - (t_{ac})_i$
For a permanent working phase, storage or dormant		ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile.

Mathematical formula for π_t

$$\pi_t = e^{2900 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)} \text{ with } t_A : \text{ambient temperature}$$

* NOTE The following conditions must be respected to obtain field values in conformance with those calculated with the above formulas:

- For **radio interference suppression** capacitors the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to **0,8**
- For other **non radio interference suppression** specified capacitors, this same ratio must be less or equal to **0,2**

with: peak voltage = continuous voltage + peak value of the alternative voltage

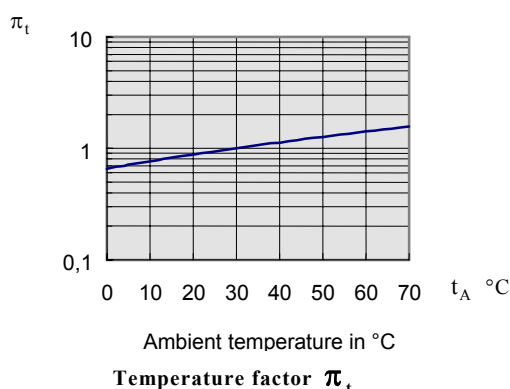
10.2 Fixed ceramic dielectric capacitors – Defined temperature coefficient – Class I (IEC 60384)

MATHEMATICAL MODEL

$$\lambda = 0.05 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 3.3 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- t_{ae}_i : average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.
 τ_i : i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
Continuous voltage applied to the capacitor See note*
Peak value of the alternative voltage applied to the capacitor See note*
Rated voltage of the capacitor. See note*



Failure modes	
Short-circuits	70 %
Open-circuits	10 %
Drifts	20 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile.	

Mathematical formula for π_t
$\pi_t = e^{1160 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

* NOTE The following condition must be respected, to obtain field values in conformance with those calculated with the above formulas:

the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to **0.5**

with: peak voltage = continuous voltage + peak value of the alternative voltage

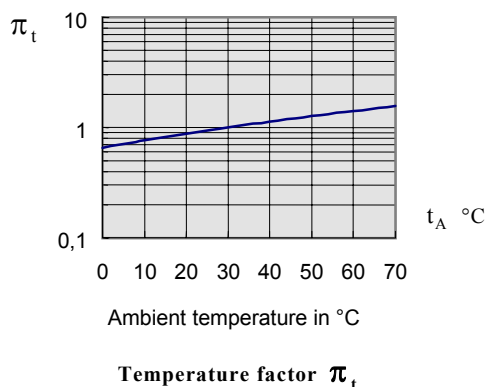
10.3 Fixed ceramic dielectric capacitors – Non defined temperature coefficient – Class II – Radio interference suppression capacitors

MATHEMATICAL MODEL

$$\lambda = 0.15 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] + 3.3 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.
 τ_i : i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
 Continuous voltage applied to the capacitor See note*
 Peak value of the alternative voltage applied to the capacitor See note*
 Rated voltage of the capacitor. See note*



Failure modes	Other ceramic, class II capacitors	Radio interference suppression capacitors
Short-circuits	90 %	70 %
Open-circuits	10 %	30 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase		$\Delta T_i = (t_{ac})_i - (t_{ae})_i$
For a permanent working phase, storage or dormant		ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile.

Mathematical formula for π_t
$\pi_t = e^{\frac{1160}{303} \left(\frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

* NOTE The following conditions must be respected, to obtain field values in conformance with those calculated with the above formulas:

- For **radio interference suppression** capacitors the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to **0,8**
- For other **non radio interference suppression** specified capacitors, this same ratio must be less or equal to **0,5**

with: peak voltage = continuous voltage + peak value of the alternative voltage

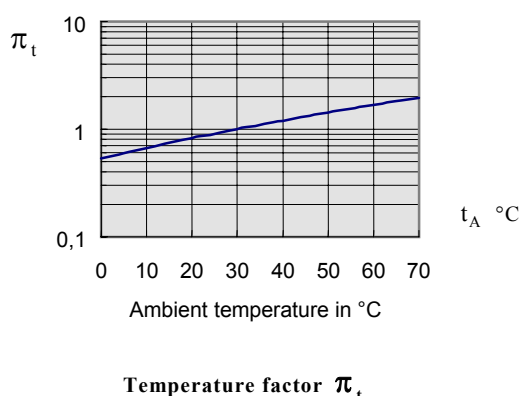
10.4 Tantalum capacitors, solid electrolyte (IEC 60384)

MATHEMATICAL MODEL

$$\lambda = 0.4 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 3.8 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

$(t_{ae})_i$	: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.	
$(t_{ac})_i$	: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.	
$(\pi_t)_i$	: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.	
τ_i	: i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.	
τ_{on}	: total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$	
τ_{off}	: time ratio for the capacitor being in storage (or dormant) mode.	
$(\pi_n)_i$	: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .	
ΔT_i	: i^{th} thermal amplitude variation of the mission profile.	
Continuous voltage applied to the capacitor		See note 1
Peak value of the alternative voltage applied to the capacitor		See note 1
Rated voltage of the capacitor.		See note 1
Situation of the capacitor		See note 2



Failure modes	
Short-circuits	80 %
Open-circuits	20 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

Mathematical formula for π_t

$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)} \text{ with } t_A : \text{ambient temperature}$$

NOTE 1 The following condition must be respected, to obtain field values in conformance with those calculated with the above formulas:

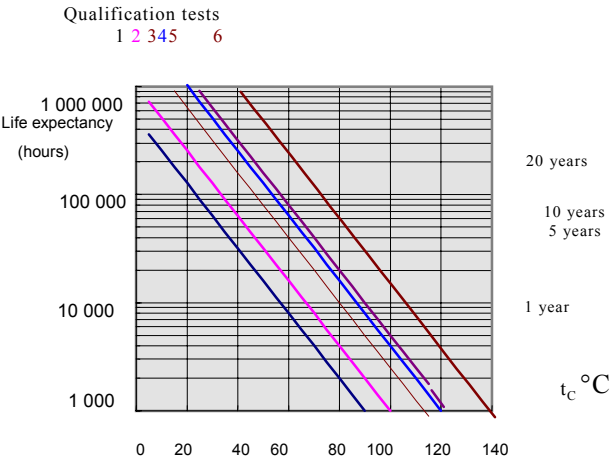
$$\text{the ratio } \frac{\text{peak voltage}}{\text{rated voltage}} \text{ must be less or equal to } 0,5$$

with: peak voltage = continuous voltage + peak value of the alternative voltage

NOTE 2 Caution must be taken against the risk of fire if the current is not limited (a fuse might be placed serially in one of the branches, between which the tantalum capacitor is positioned).

10.5 Aluminum, non-solid electrolyte capacitors - Life expectancy

Failure rate of non solid electrolyte aluminum capacitors is assumed to be constant but only within the life expectancy limits.



Qualification test conditions (duration , temperature) According to capacitor types		
Qualification tests	Duration (h)	Temperature (°C)
1	1 000	85
2	2 000	85
3	5 000	85
4	2 000	105
5	10 000	85
6	2 000	125

Life expectancy according to qualifications
Test conditions of the capacitor

With: t_c : temperature of the capacitor, default value
 $t_c = t_{\text{ambient}} + 5^\circ\text{C}$

*NOTE 1 Life expectancy depends on the operating temperature and on the qualification test conditions (i.e. on the technology type)
*NOTE 2 Ambient temperature for a non solid electrolyte aluminum capacitor is taken at its immediate vicinity (overheating of the other components has to be taken into account).

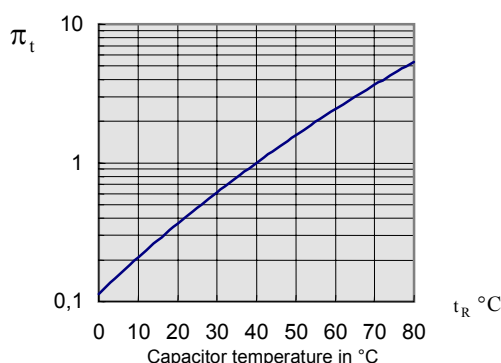
Mathematical formula for the life expectancy
$\text{Life expectancy (hours)} = \text{qualification test duration} \times 2^{\left(\frac{(T_M+5)-T_c}{10}\right)}$ with T_M : maximum temperature of the climatic category

MATHEMATICAL MODEL

$$\lambda = 1.3 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] \times \pi_A + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.
 τ_i : i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.
 π_A : factor related to the self heating
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
Continuous voltage applied to the capacitor See note
Rated voltage of the capacitor. See note
Applied ripple current
Maximum admissible ripple current.



Temperature factor π_t

Mathematical formula for π_t

$$\pi_t = e^{4640 \left(\frac{1}{313} - \frac{1}{273+t_R} \right)} \text{ with } t_R : \text{capacitor temperature}$$

Capacitor temperature t_R °C:

CAUTION !

The capacitor temperature is different to that of the ambient temperature if a current flows through the capacitor (particularly pulses for example) or if it is heated by a radiative environment (a dissipating component). It is necessary to:

- measure the case temperature,
- or, by default

⇒ take into account the heat dissipated by the radiative environment;

⇒ take into account the temperature rise Δt due to the current through the capacitor (self heating) by:

$$\Delta t = 20 \left(\frac{\text{Applied ripple current}}{\text{Maximum admissible current}} \right)^2$$

CAUTION !

Life expectancy of non solid electrolyte, aluminum capacitors is limited and very sensible to ambient temperature

Failure modes

Rated voltage	< 350 V	≥ 350 V
Short circuits	30 %	50 %
Open circuits	30 %	0 %
Drifts	40 %	50 %

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$

n_i : Annual number of cycles with the amplitude ΔT_i

For an on/off phase	$\Delta T_i = \left[\frac{T_R}{3} + (t_{ac})_i \right] - (t_{ae})_i$
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile

Peak value of the pulsed current

Ratio	π_A
≤ 1,5	1
1,5 to 2	3
2 to 3	10

NOTE The following condition must be respected, to obtain field values in conformance with those calculated with the above formulas:

the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to 0,5

with: peak voltage = continuous voltage + peak value of the alternative voltage

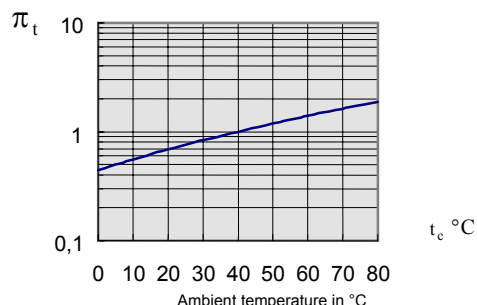
10.6 Aluminum electrolytic capacitor, solid electrolyte

MATHEMATICAL MODEL

$$\lambda = 2.4 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^i (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.
 τ_i : i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
 Continuous voltage applied to the capacitor See note
 Peak value of the alternative voltage applied to the capacitor See note
 Rated voltage of the capacitor. See note



Failure modes	
Short-circuits	10 %
Open-circuits	90 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase		$\Delta T_i = (t_{ac})_i - (t_{ac})_i$
For a permanent working phase, storage or dormant		ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile

Mathematical formula for π_t
$\pi_t = e^{1740 \left(\frac{1}{313} - \frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

NOTE The following condition must be respected, to obtain field values in conformance with those calculated with the above formulas:

the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to **0,8**

with: peak voltage = continuous voltage + peak value of the alternative voltage

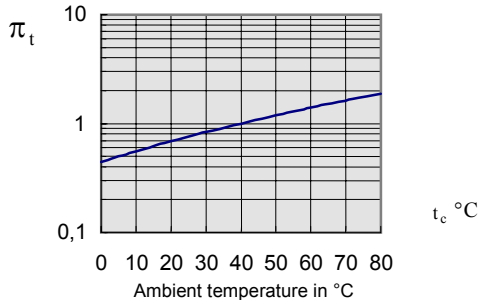
10.7 Aluminum electrolytic capacitor, polymer electrolyte (IEC 60384)

MATHEMATICAL MODEL

$$\lambda = 0.6 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $t_{ae)i}$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ae})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.
 τ_i : i^{th} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
Continuous voltage applied to the capacitor See note
Peak value of the alternative voltage applied to the capacitor See note
Rated voltage of the capacitor. See note



Temperature factor π_t

Failure modes	
Short-circuits	10 %
Open-circuits	90 %

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

Mathematical formula for π_t
$\pi_t = e^{1740 \left(\frac{1}{313} - \frac{1}{273 + t_A} \right)}$ with t_A : ambient temperature

NOTE The following condition must be respected, to obtain field values in conformance with those calculated with the above formulas:

the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to **0,8**

with: peak voltage = continuous voltage + peak value of the alternative voltage

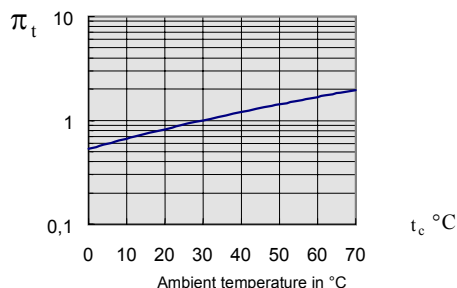
10.8 Variable ceramic capacitors, disks (Dielectric ceramic) (IEC 60384)

MATHEMATICAL MODEL

$$\lambda = 0.16 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 3.3 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / \text{h}$$

NECESSARY INFORMATION:

- $(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the capacitor mission profile.
 τ_i : i^{eme} working time ratio of the capacitor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the capacitor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the capacitor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
 Continuous voltage applied to the capacitor See note
 Peak value of the alternative voltage applied to the capacitor See note
 Rated voltage of the capacitor. See note



Temperature model π_t

Mathematical formula for π_t

$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)} \text{ with } t_A : \text{ambient temperature}$$

Failure modes

Short-circuits	40 %
Open-circuits	10 %
Drifts	50 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

NOTE: The following condition must be respected, to obtain field values in conformance with those calculated with the above formulas:

the ratio $\frac{\text{peak voltage}}{\text{rated voltage}}$ must be less or equal to **0,5**

with: peak voltage = continuous voltage + peak value of the alternative voltage

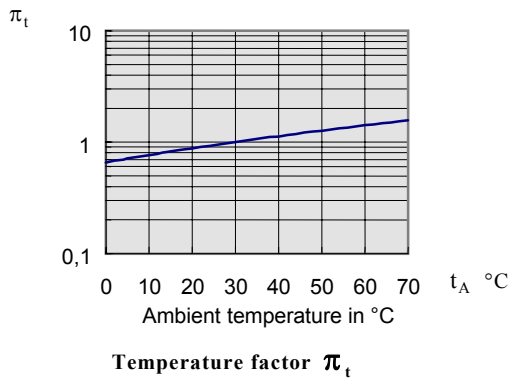
10.9 Thermistors with negative temperature coefficient (NTC) (IEC 60539)

MATHEMATICAL MODEL

$$\lambda = 3 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $t_{ae)i}$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the thermistor mission profile.
 τ_i : i^{th} working time ratio of the thermistor for the $i^{ème}$ junction temperature of the mission profile.
 τ_{on} : total working time ratio of the thermistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the thermistor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the package, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.



Mathematical formula for π_t

$$\pi_t = e^{1160 \left(\frac{1}{303} - \frac{1}{273 + t_A} \right)} \text{ with } t_A : \text{ambient temperature}$$

Répartition des défauts

Short-circuits	70 %
Open-circuits	10 %
Drifts	20 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase		$\Delta T_i = (t_{ac})_i - (t_{ae})_i$
For a permanent working phase, storage or dormant		ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile

11 Resistors and potentiometers (IEC 60115)

11.1 Fixed, low dissipation film resistors – High stability (rs), general purpose (rc), “minimelf”

MATHEMATICAL MODEL

$$\lambda = 0.1 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right] \times 10^{-9} / \text{h}$$

NECESSARY INFORMATION:

$(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.

τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

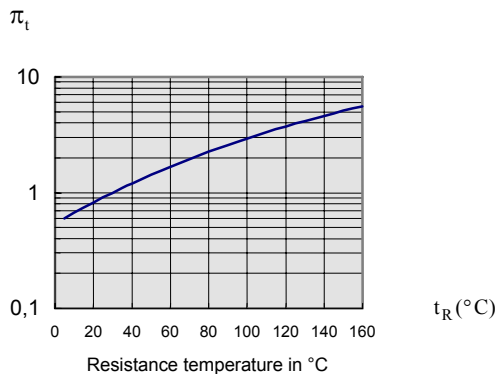
τ_{off} : time ratio for the resistor being in storage (or dormant) mode.

$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

Power applied to the resistor.

Rated power of the resistor.



Failure modes	
Open-circuits	40 %
Drifts	60 %

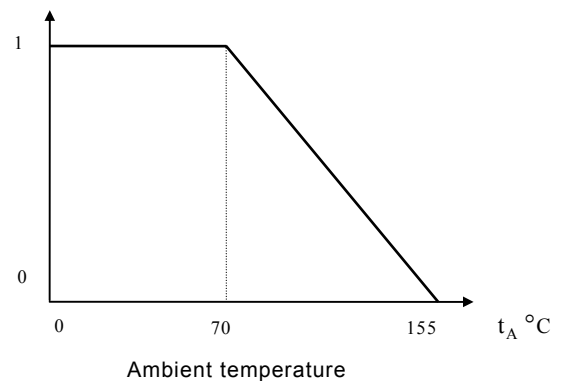
Mathematical formula for π_t	
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_R} \right)}$	with
	t_R = Resistor temperature
	t_A = Ambient temperature
	$t_R = t_A + 85 \times \frac{\text{Operating power}}{\text{Rated power}}$

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

For information:

Maximum operating power (derating curve)

Ratio: $\frac{\text{Maximum operating power}}{\text{Rated power}}$



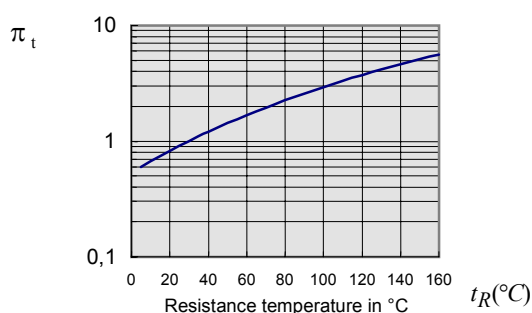
11.2 Hot molded carbon composition, fixed resistors (IEC 60115)

MATHEMATICAL MODEL

$$\lambda = 0.5 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $(t_{ae})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.
 τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the resistor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
Power applied to the resistor.
Rated power of the resistor.



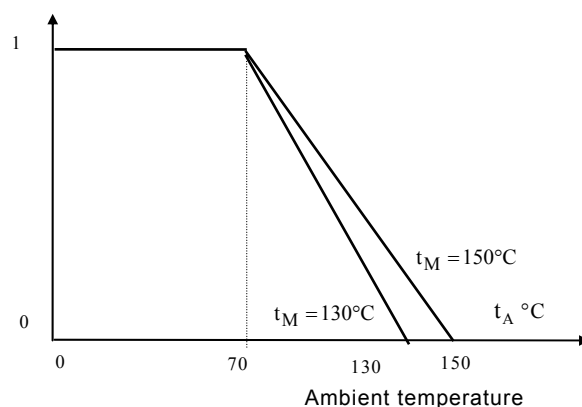
Failure modes	
Short-circuits	0 %
Open-circuits	100 %

Mathematical formula for π_t	
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_R} \right)}$	with
	t_R = Resistance temperature
	t_A = Ambient temperature
	t_M = maximum rated temperature
For $t_M = 130^\circ\text{C}$	
$t_R = t_A + 60 \times \frac{\text{Operating power}}{\text{Rated power}}$	
For $t_M = 150^\circ\text{C}$	
$t_R = t_A + 80 \times \frac{\text{Operating power}}{\text{Rated power}}$	

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

For information:
Maximum operating power (derating curve)

Ratio: $\frac{\text{Maximum operating power}}{\text{Rated power}}$



11.3 Fixed, high dissipation film resistors (IEC 60115)

MATHEMATICAL MODEL

$$\lambda = 0.4 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

t_{ae}_i : average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.

τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

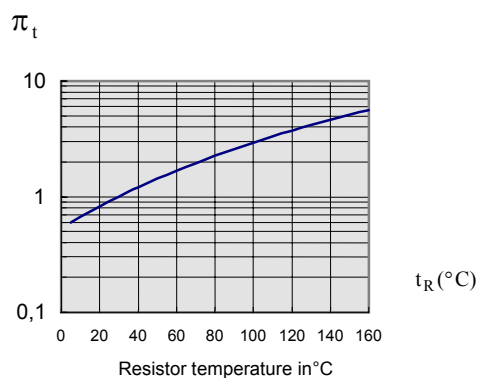
τ_{off} : time ratio for the resistor being in storage (or dormant) mode.

$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

Power applied to the resistor.

Rated power of the resistor.



Failure modes	
Short-circuits	0 %
Open-circuits	100 %

Mathematical formula for π_t

with

$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273 + t_R} \right)}$$

t_R = Resistor temperature
 t_A = Ambient temperature

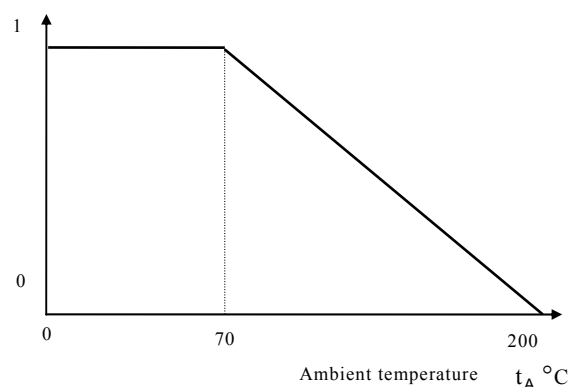
$$t_R = t_A + 130 \times \frac{\text{Operating power}}{\text{Rated power}}$$

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[(t_{ac})_i + \frac{t_R}{3} \right] - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

For information:

Maximum operating power (derating curve)

Ratio: $\frac{\text{Maximum operating power}}{\text{Rated power}}$



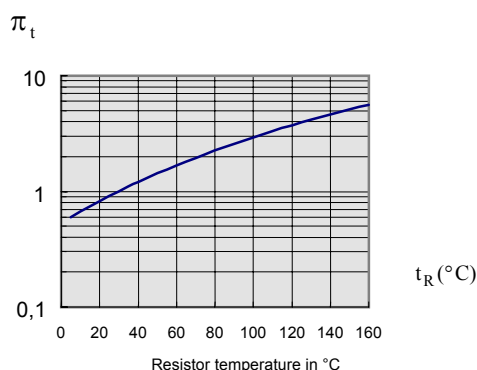
11.4 Low dissipation wirewound resistors (IEC 60115)

MATHEMATICAL MODEL

$$\lambda = 0.3 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- t_{ae}_i : average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.
 τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the resistor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
Power applied to the resistor.
Rated power of the resistor.



Failure modes	
Short-circuits	0 %
Open-circuits	100 %

Mathematical formula for π_t

with

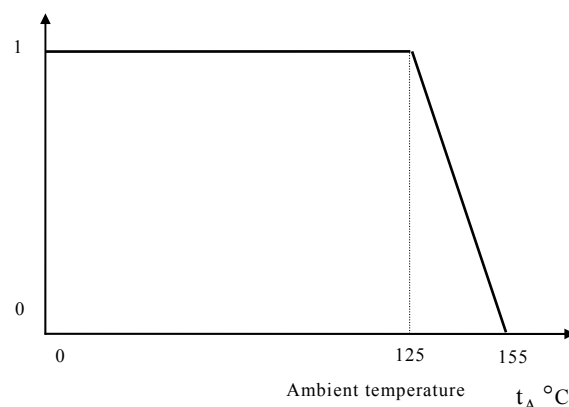
$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273 + t_R} \right)}$$

t_R = Resistor temperature
 t_A = Ambient temperature
 $t_R = t_A + 30 \times \frac{\text{Operating power}}{\text{Rated power}}$

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

For information:
Maximum operating power (derating curve)

Ratio: $\frac{\text{Maximum operating power}}{\text{Rated power}}$



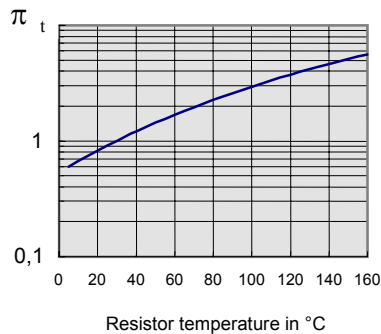
11.5 High dissipation wirewound resistors (IEC 60115)

MATHEMATICAL MODEL

$$\lambda = 0.4 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] + 1.4 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

- t_{ae}_i : average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.
 τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the resistor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
 Power applied to the resistor.
 Rated power of the resistor.



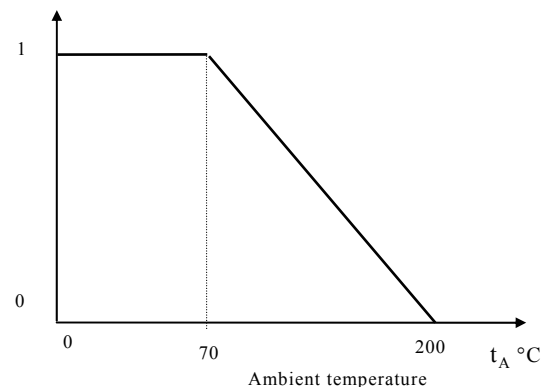
Failure modes	
Short-circuits	0 %
Open-circuits	100 %

Mathematical formula for π_t	
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273 + t_R} \right)}$	with
	t_R = Resistor temperature
	t_A = Ambient temperature
$t_R = t_A + 130 \times \frac{\text{Operating power}}{\text{Rated power}}$	

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = \left[(t_{ac})_i + \frac{t_R}{3} \right] - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

For information:
Maximum operating power (derating curve)

Ratio: $\frac{\text{Maximum operating power}}{\text{Rated power}}$



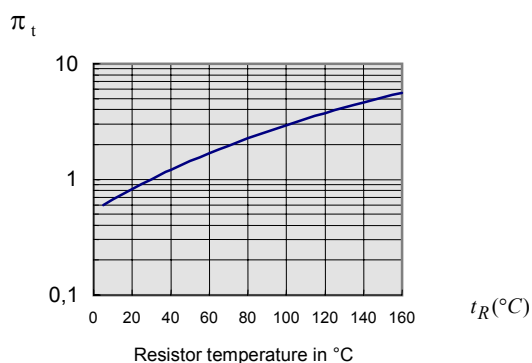
11.6 Fixed, low dissipation surface mounting resistors and resistive array (IEC 60115)

MATHEMATICAL MODEL

$$\lambda = 0.01 \times \left[\frac{\sum_{i=1}^y (\pi_i) \times \tau_i}{\tau_{on} + \tau_{off}} \right] \times \sqrt{N} + 3.3 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

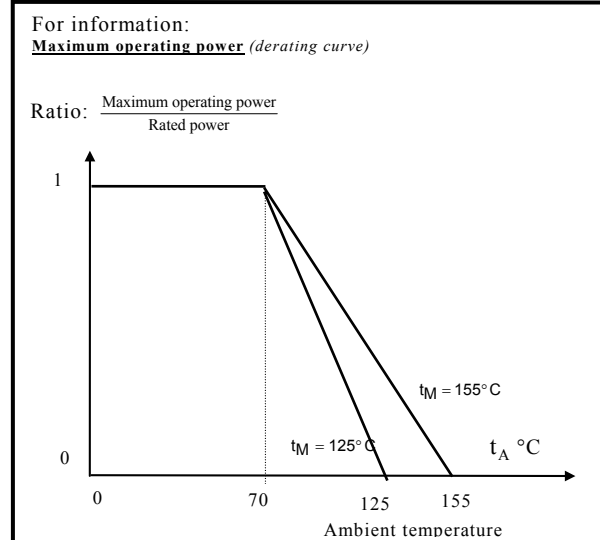
- $t_{ae,i}$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ae})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 $(\pi_i)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.
 τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the resistor being in storage (or dormant) mode.
 N : resistors number of the resistive array.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.
Power applied to the resistor.
Rated power of the resistor..



Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

Mathematical formula for π_t	
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_R} \right)}$	with
	t_R = resistor temperature
	t_A = Ambient temperature
	t_M = Maximum rated category ,
for $t_M = 125^\circ \text{C}$	
$t_R = t_A + 55 \times \frac{\text{Operating power}}{\text{Rated power}}$	
for $t_M = 155^\circ \text{C}$	
$t_R = t_A + 85 \times \frac{\text{Operating power}}{\text{Rated power}}$	

Failure modes	
Drifts	60 %
Open-circuits	40 %



11.7 Non wirewound cermet potentiometer (one or several turn) (IEC 60393)

MATHEMATICAL MODEL

$$\lambda = 0.3 \times \left[\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right] \times \pi_Y + 1.2 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \times 10^{-9} / h$$

NECESSARY INFORMATION:

$(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the resistor mission profile.

τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

τ_{off} : time ratio for the resistor being in storage (or dormant) mode.

π_Y : factor related to the annual number of shaft rotation.

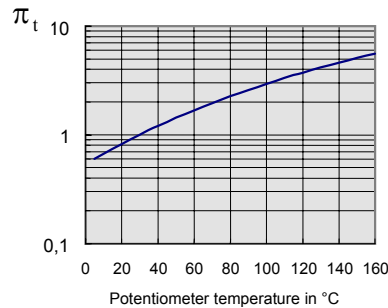
$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

R_L and R_P for π_W calculation.

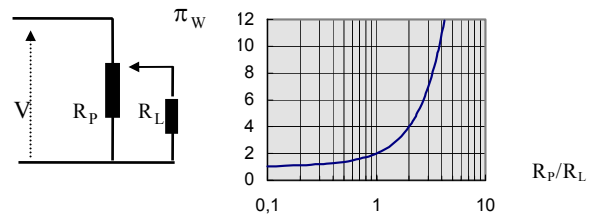
Power applied to the resistor.

Rated power of the resistor.



Failure modes	
Open-circuits	80 %
Drifts	20 %

Number of annual shaft rotations		π_Y
≤ 10		0,2
> 10		1



Mathematical formulas for π_t , and π_W

with

t_R = potentiometer temperature

t_A = ambient temperature

t_M = maximum rated temperature ,

for $t_M = 125^\circ\text{C}$

$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_R} \right)}$$

and

$$\pi_W = 1 + \frac{1}{2} \left[\frac{R_P^2}{4R_L^2} + \frac{R_P}{R_L} \right]$$

for $t_M = 155^\circ\text{C}$

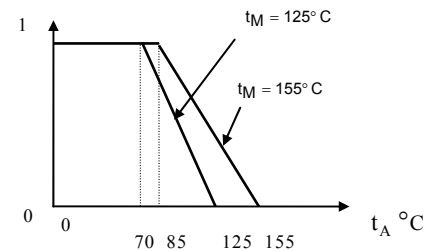
$$t_R = t_A + 55 \times \frac{\frac{V^2}{R_P} \times \pi_W}{\text{Rated power}}$$

$$t_R = t_A + 70 \times \frac{\frac{V^2}{R_P} \times \pi_W}{\text{Rated power}}$$

For information:

Maximum operating power (derating curve)

Ratio: $\frac{\text{Maximum operating power}}{\text{Rated power}}$



Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

12 Inductors and transformers (IEC 61248)

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \left(\frac{\sum_{i=1}^y (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} + 7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

NECESSARY INFORMATION:

$(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.

$(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.

$(\pi_t)_i$: i^{th} temperature factor related to the i^{th} junction temperature of the component mission profile.

τ_i : i^{th} working time ratio of the component for the i^{th} junction temperature of the mission profile.

τ_{on} : total working time ratio of the component. With: $\tau_{on} = \sum_{i=1}^y \tau_i$

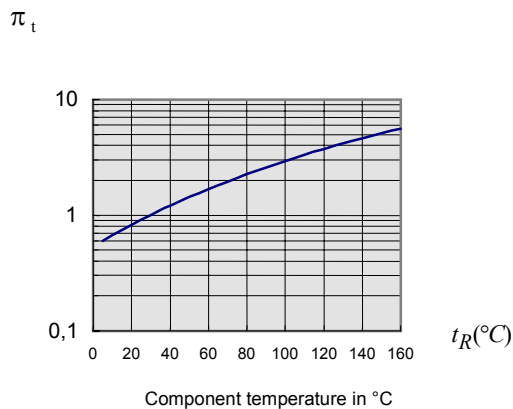
τ_{off} : time ratio for the component being in storage (or dormant) mode.

$(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the component, with the amplitude ΔT_i .

ΔT_i : i^{th} thermal amplitude variation of the mission profile.

Power loss by the component.

Radiating surface of the component.



Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase of a power inductor	$\Delta T_i = \left[(t_{ac})_i + \frac{t_R}{3} \right] - (t_{ac})_i$	
For an on/off phase of a low current inductor	$\Delta T_i = (t_{ac})_i - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

Mathematical formula for π_t	
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_R} \right)}$	with: t_R = component temperature t_A = ambient temperature around component $t_R = t_A + 8.2 \times \frac{\text{Power loss (watt)}}{\text{Radiating surface (dm}^2\text{)}}$

Failure rates according to the component type (FIT)			λ_0
Inductors	Low current inductors	fixed	0,2
		variable	0,4
	Power inductors (50 Hz, chopping, filtering)		0,6
Transformers	Signal transformers		1,5
	Power transformers		3

Failure modes	
Short-circuits	20 %
Open-circuits	80 %

13 Microwave passive components, piezoelectric components and surface acoustic wave filters (IEC 61261, IEC 61019, IEC 60368)

13.1 Microwave passive components

13.2 Piezoelectric components

13.3 Surface acoustic wave filters

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \left(\left[\frac{\tau_{on}}{\tau_{on} + \tau_{off}} \right] + 3 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

NECESSARY INFORMATION:

- $(t_{ac})_i$: average outside ambient temperature surrounding the equipment, during the i^{th} phase of the mission profile.
 $(t_{ac})_i$: average ambient temperature of the printed circuit board (PCB) near the components, where the temperature gradient is cancelled.
 τ_i : i^{th} working time ratio of the resistor for the i^{th} junction temperature of the mission profile.
 τ_{on} : total working time ratio of the resistor. With: $\tau_{on} = \sum_{i=1}^y \tau_i$
 τ_{off} : time ratio for the resistor being in storage (or dormant) mode.
 $(\pi_n)_i$: i^{th} influence factor related to the annual cycles number of thermal variations seen by the resistor, with the amplitude ΔT_i .
 ΔT_i : i^{th} thermal amplitude variation of the mission profile.

Component types		λ_0 FIT
Microwave passive components	Fixed	2
	Variables	4
	With ferrite	4
Piezoelectric components	Resonators quartz or ceramic filters	5
	Oscillators: XO, PXO	10
	Oscillators: VCXO, TCXO	15
	Oscillators: OCXO	30
Acoustic wave filters		15

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase of a passive, piezoelectric or acoustic microwave component	$\Delta T_i = (t_{ac})_i - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

14 Relays

14.1 Evaluating voltage and current (v_t , i_t) in transient conditions

(Voltage at the terminals of a contact and current through the contact)

14.1.1 Preferred method

The preferred method is to measure V_t , I_t in transient conditions, or evaluate them by analysing the actual circuit diagram.

14.1.2 Default method

By default, the following methods will be used (depending on how much is known about the circuit).

14.1.2.1 Calculate or measure the voltage and current V , I in steady state conditions (voltage at the terminals, current through the contact).

14.1.2.2 Determine the Figure 2 equivalent circuit diagram from the actual circuit diagram.

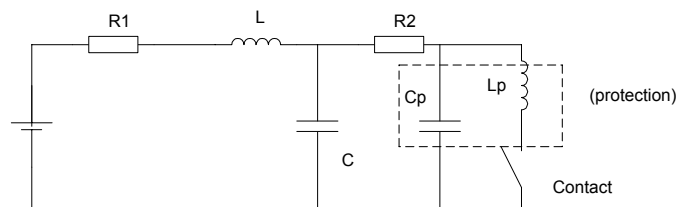


Figure 2 – Equivalent diagram representing the circuit of a relay contact

14.1.2.2.1 The inductance L is given by the sum of all series inductance values in the real contact circuit (only inductances of greater than $100\ \mu\text{H}$ are included, and protection inductance L_p is not included).

14.1.2.2.2 The value of capacitor C is the sum of the capacitance values of the capacitors across the terminals of the contact in the real circuit: only capacitors with a value of greater than $100\ \text{pF}$ and located within the area bounded by the contact and the first inductance of more than $100\ \mu\text{H}$ in the real circuit are counted (see Figure 3). The line capacitance is included ($100\ \text{pF}$ per metre). The value of the protection capacitor C_p is not included.

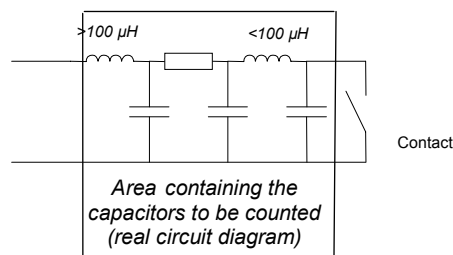


Figure 3 – Positions of capacitors in the real circuit diagram for which the values must be counted in C

14.1.2.2.3 The respective values of resistors R_1 and R_2 are the sum of the resistors found in the real circuit diagram, on the one hand in the area marked in Figure 3, and on the other hand outside the area (the resistance of the protection circuit is not included).

14.1.2.2.4 Inductor L_p and capacitor C_p are respectively the protection inductor and protection capacitor (not included when calculating L or C).

NOTE Protection diodes, if any, are not included (nor is the associated protection resistor, as stated in 14.1.2.2.3).

14.1.2.3 Calculate the voltage V_t and current I_t in transient conditions.

14.1.2.3.1 The tables in Figures 5, 6 and 7 give ratios V_t/V and I_t/I (V , I in steady-state conditions; V_t , I_t in transient condition) according to how much is known about the contact circuit. For greater clarity, the tables are given according to circuit type (resistive, capacitive, inductive, inductive and capacitive), summarized as shown in Figure 4, in other words in the four regions defined by the values of inductors L and capacitors C . The four regions are bounded by the characteristic values of L ($100 \mu\text{H}$) and C (1 nF).

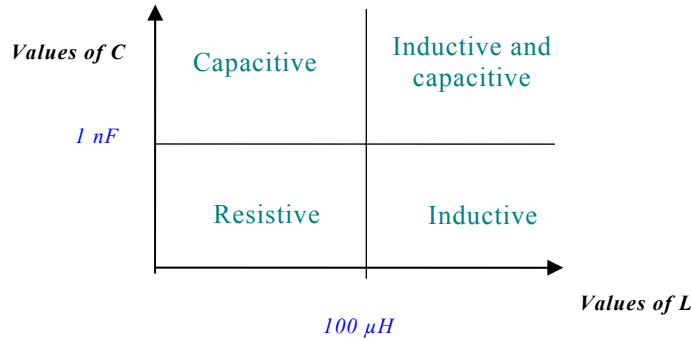


Figure 4 – Regions adopted for the purposes of Figures 5, 6 and 7

14.1.2.3.2 Results

A. Case 1: values of L , C and R are known (respectively in mH , nF and $\text{k}\Omega$), with $R = R_1 + R_2$.

The equations which give the transient values V_t , I_t for voltage and current, which are needed to calculate lifetime θ and pollution factor π_p , are given in the Tables in Figure 5 according to values of L , C , R ($R = R_1 + R_2$), C_p and L_p , which are assumed known. What is in fact given are the ratios V_t/V and I_t/I , with V and I being values in steady-state conditions for voltage at the terminals with the contact open and current through the contact).

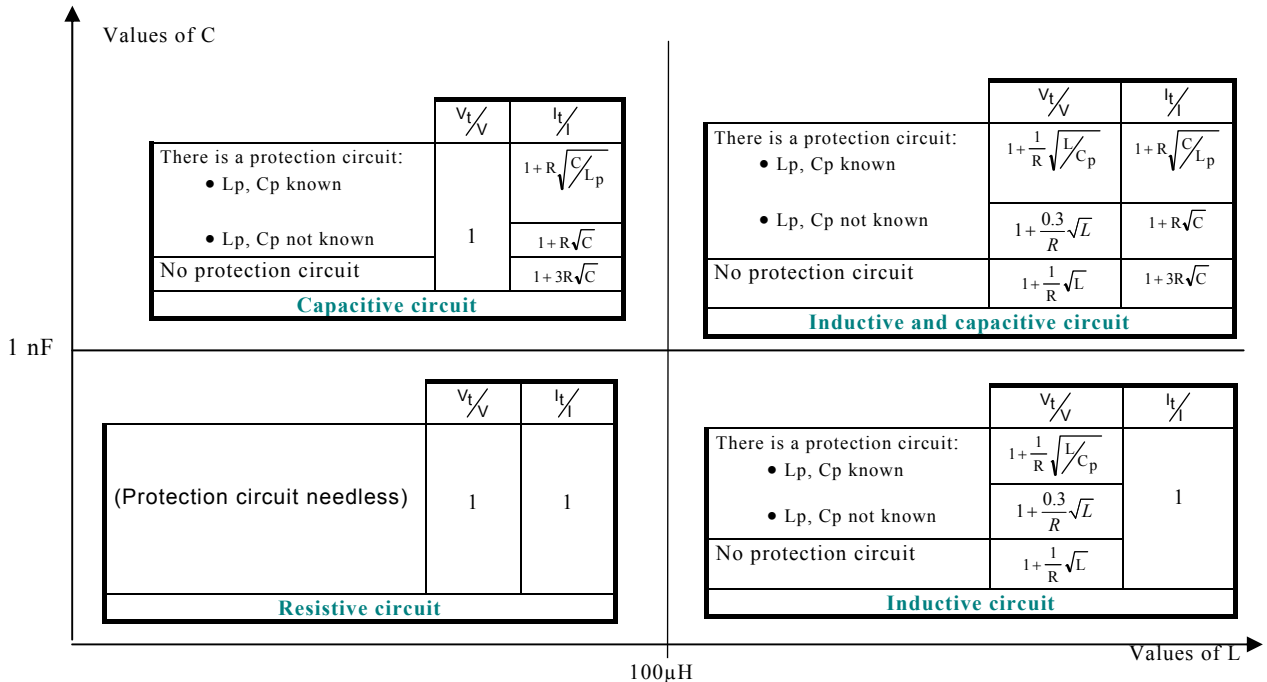


Figure 5 – Evaluating the ratios V_t/V and I_t/I according to $R = R_1 + R_2$, C , L , and C_p , L_p (R in $\text{k}\Omega$, C , C_p in nF ; L , L_p in mH)

B. Case 2: values of L, C and R are not known, but the equivalent diagram has been constructed and the nature of the circuit - capacitive or inductive and capacitive - is known.

The Tables in Figure 6 give ratios V_t/V and I_t/I according to how much is known about L_p , C_p and depending on whether the life expectancy or failure rate is being calculated.

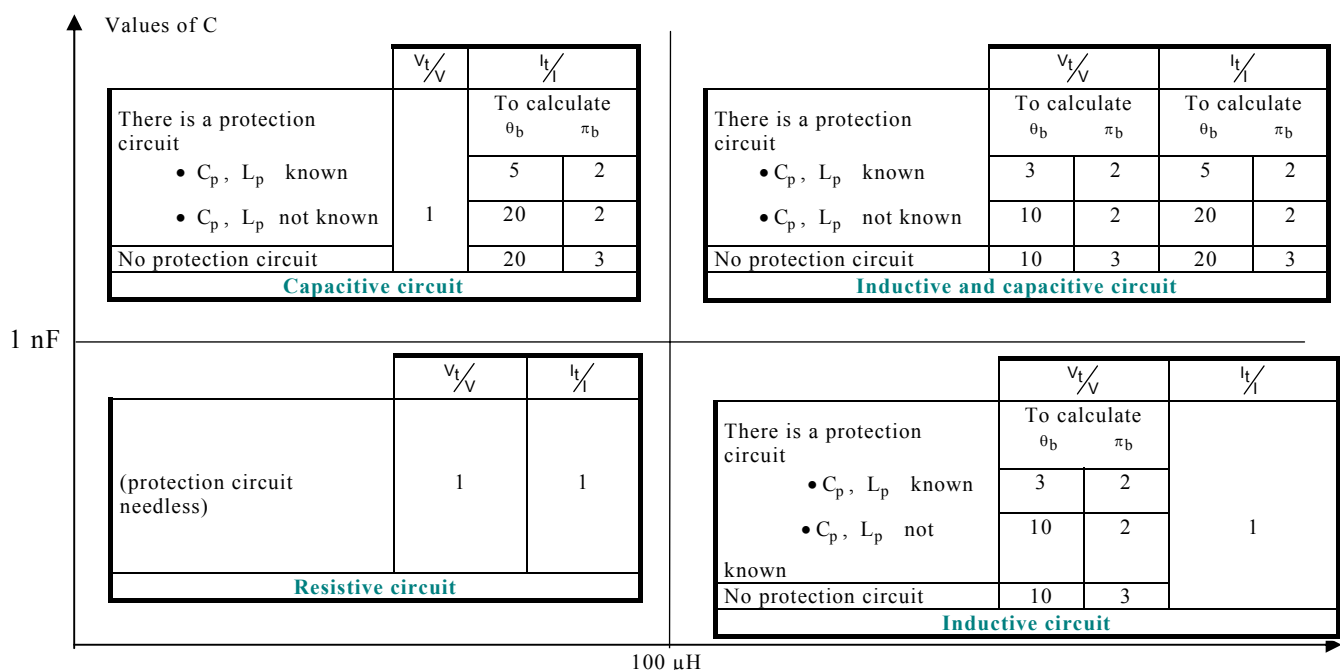


Figure 6 – Evaluating ratios V_t/V and I_t/I when L and C are not known

C. Case 3: Nothing is known about the electrical circuit of the contact, the Table in Figure 7 gives default values for the ratios V_t/V and I_t/I (depending whether life expectancy θ_b or failure rate (π_p factor) is being calculated).

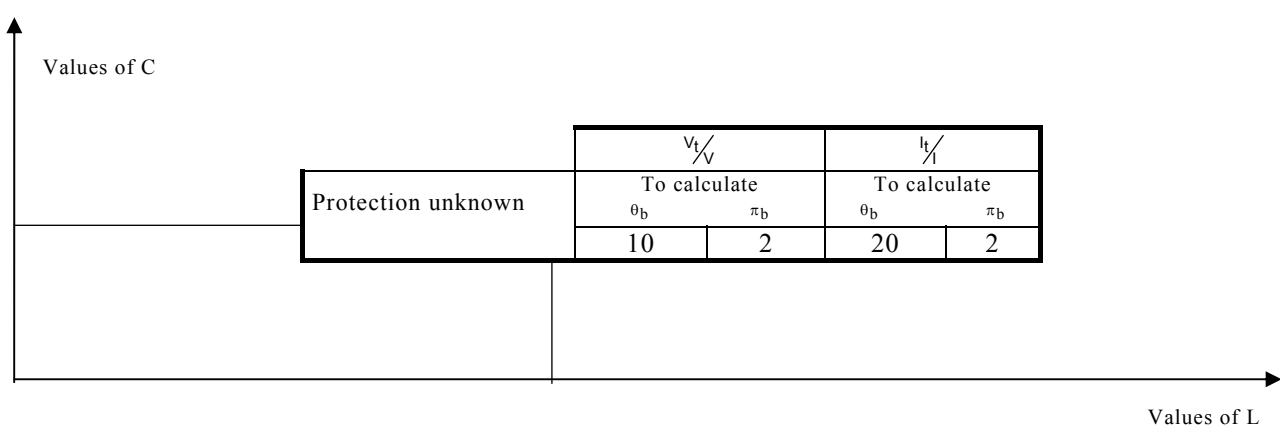


Figure 7 – Default values of V_t/V and I_t/I when nothing is known about the electrical circuit of the contact

14.2 Mercury wetted reed relays, low power (IEC 60255)

Balanced – latching balanced (power up to 250 W)

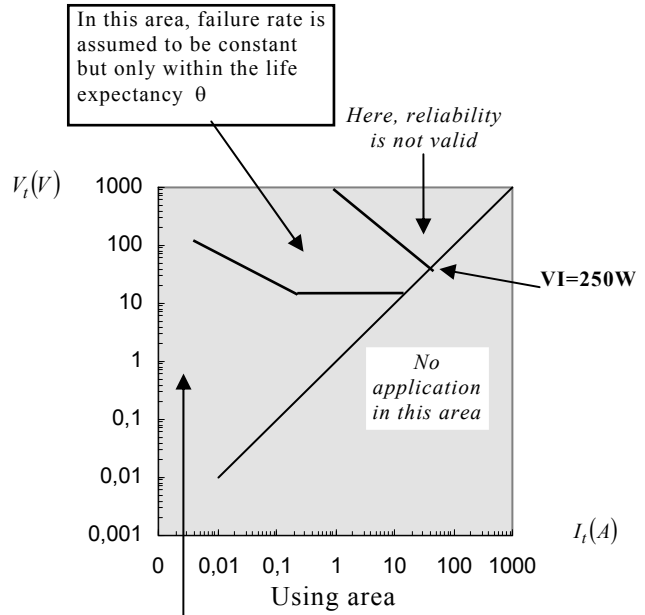
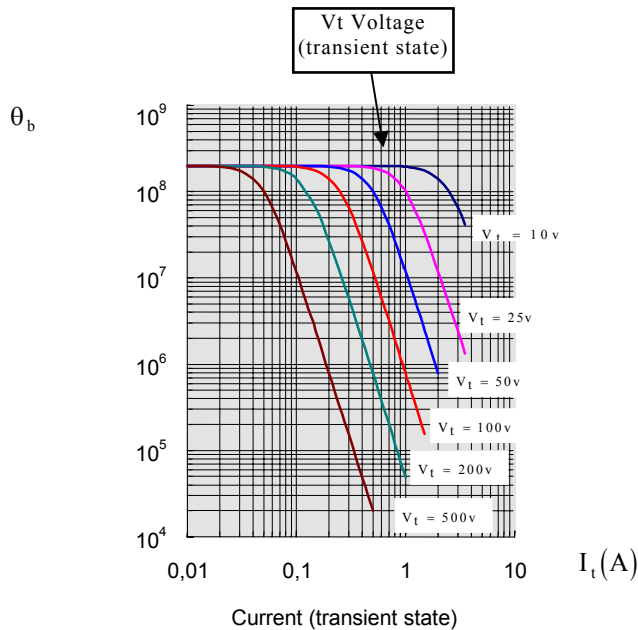
LIFE EXPECTANCY: θ

$$\theta = \theta_b \frac{1}{\pi_t \times N} \text{ in h}$$

- π_t is given in next page (same factor as λ for the failure rate).
- N: Number of cycles per hour

Mathematical formula for θ_b

$$\theta_b = \frac{2 \times 10^8}{1 + \left(\frac{V_t \times I_t}{25} \right)^4}$$



In this area, failure rate is assumed to be constant during unlimited time.

The voltage V_t and the current I_t values (transient state) are given on pages 74 to 76. (Select the appropriate column " θ_b calculation").

θ_b depending on the voltage V_t and the current I_t in transient state (see 14.1)

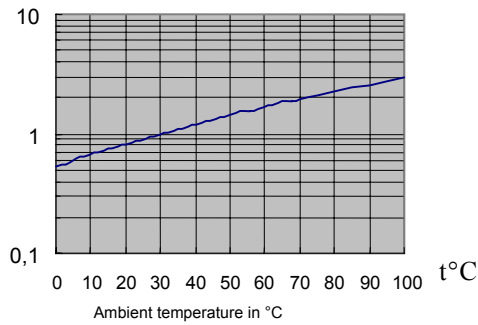
MATHEMATICAL MODEL

$$\lambda = 0.3 \times \pi_t \times \pi_T \times \pi_Y \times \pi_C \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

with $\pi_C = \pi_{C1}$ OR π_{C2}

CAUTION !

Life expectancy is limited !

 π_t


Relay type	π_T
Monostable	1
Bistable latching	5

Operating cycles per hours: mean number N	π_Y
$N \leq 10$	1
$N > 10$	$\sqrt{\frac{N}{10}}$

Mathematical formula for π_t
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

Failure modes	
Short-circuits	50 %
Open-circuits	50 %

Necessary information	For
Current, voltage	θ
Load circuit	θ
Number of active contacts	π_C
Operating cycles per hours: mean number N	π_Y
Ambient temperature t_A	π_t
Relay type	π_T

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

Number of breaker active contacts "break contact" or "make contact" (form A and B)	π_{C1}
No contact of this type	0
1	1
2	1,5
3	2
4	2,5

Number of inverter active contact "break contact" and "make contact" (form C)	π_{C2}
No contact of this type	0
1	1,8
2	3
3	4,3
4	5,5
6	8

14.3 Dry reed relays (IEC 60255)

Balanced – Latching balanced (power up to 150 W)

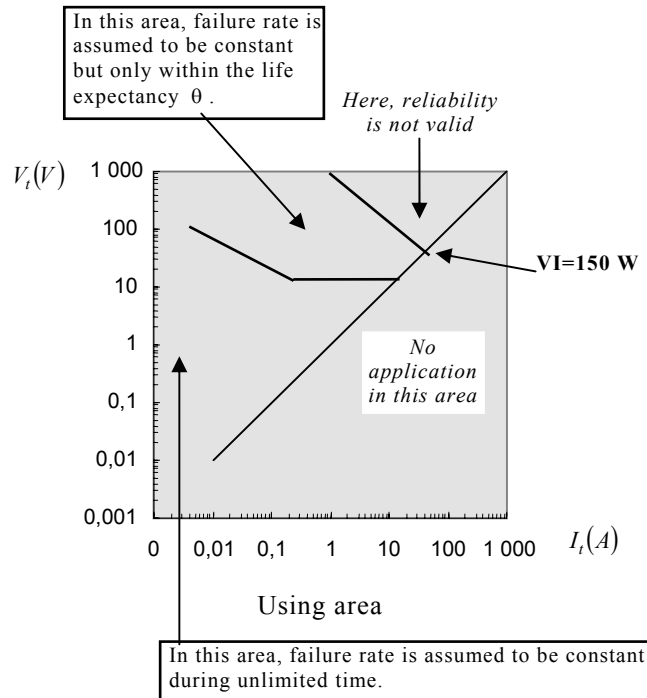
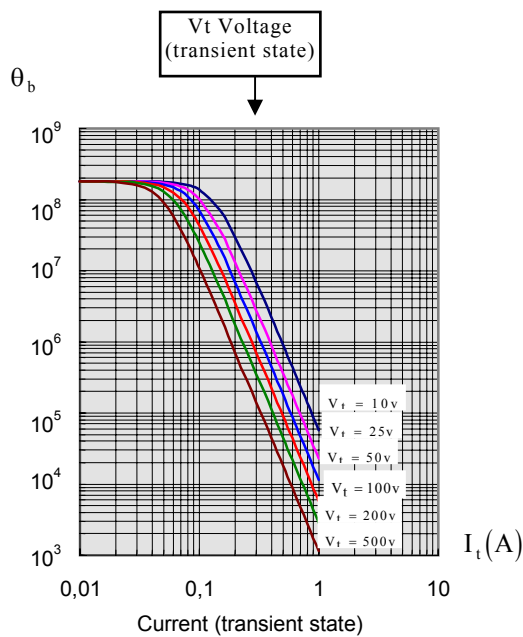
LIFE EXPECTANCY

$$\theta = \theta_b \frac{1}{\pi_t \times N} \text{ in h}$$

- π_t is given in next page (same factor as λ for the failure rate).
- N: Number of cycles per hour

Mathematical formula for θ_b

$$\theta_b = \frac{1.8 \times 10^8}{1 + \left(\frac{V_t}{2}\right) \times \left(\frac{I_t}{0,2}\right)^4}$$



The voltage V_t and the current I_t values (transient state) are given in 14.1 (Select the appropriate column " θ_b calculation").

θ_b depending on the voltage V_t and the current I_t in transient state (see 14.1)

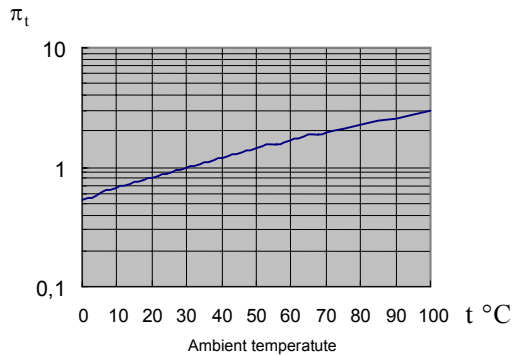
MATHEMATICAL MODEL

$$\lambda = 0.75 \times \pi_t \times \pi_T \times \pi_p \times \pi_Y \times \pi_C \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

with $\pi_C = \pi_{C1}$ or π_{C2}

CAUTION !

Life expectancy is limited!



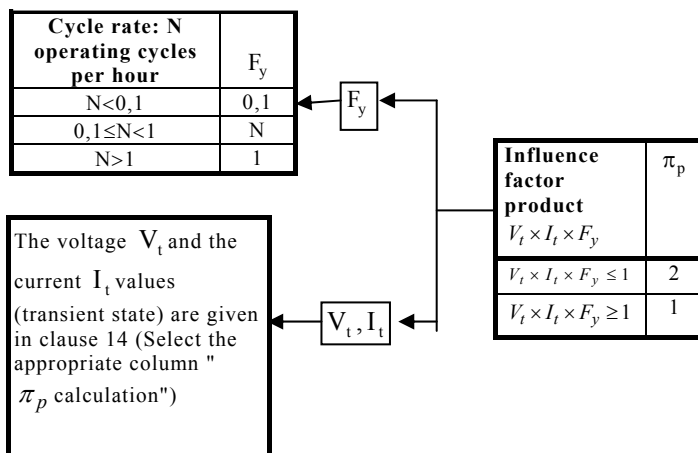
Relay type	π_T
Monostable	1
Bistable latching	5

Operating cycles per hours: mean number N	π_Y
$N \leq 10$	1
$N > 10$	$\sqrt{\frac{N}{10}}$

Mathematical formula for π_t
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273 + t_A} \right)}$ with t_A : ambient temperature

Pollution factor π_p depending on the operating cycle rate.

Voltage V_t and current I_t (transient state)



Necessary information	for
Current, voltage	θ
Load circuit	θ
Number of active contacts	π_C
Operating cycles per hours: mean number N	π_Y, F_y
Ambient temperature t_A	π_t
Relay type	π_T
Environment	π_p

Number of breaker active contacts "break contact" or "make contact" (form A and B)	π_{C1}
No contact of this type	0
1	1
2	1,5
3	2
4	2,5

Number of inverter active contact "break contact" and "make contact" (form C)	π_{C2}
No contact of this type	0
1	1,8
2	3
3	4,3
4	5,5
6	8

Failure modes	
Short-circuits	50 %
Open-circuits	50 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.6}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

14.4 Electromechanical relays, miniature or card – European type, thermal relays (power up to 500 W) (IEC 60255)

LIFE EXPECTANCY θ

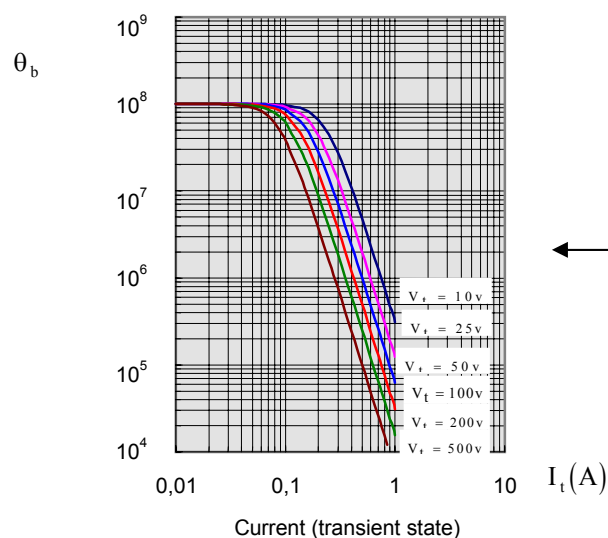
$$\theta = \theta_b \frac{1}{\pi_t \times N} \quad \text{in h}$$

- π_t is given in next page (same factor as λ for the failure rate).
- N: Number of cycles per hour

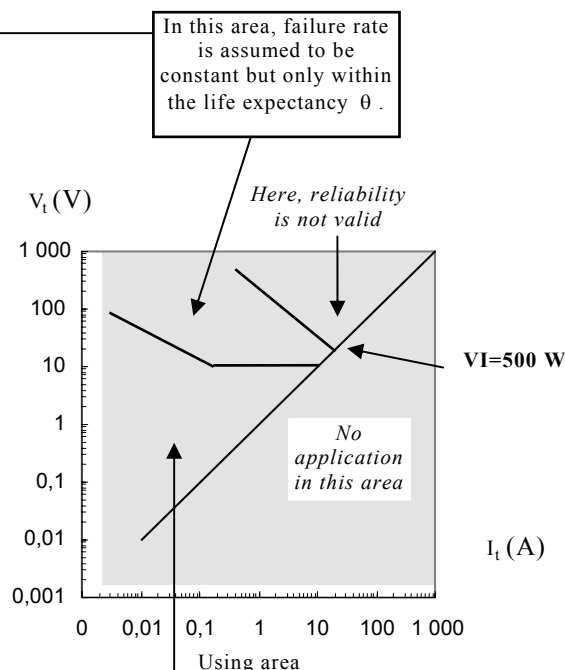
Mathematical formula for θ_b

$$\theta_b = \frac{10^8}{1 + 2V_t \times \left(\frac{I_t}{0.5}\right)^4}$$

V_t Voltage
(transient state)



θ_b depending on the voltage V_t and the current I_t in transient state (see 14.1)



In this area, failure rate is assumed to be constant during unlimited time.

The voltage V_t and the current I_t values (transient state) are given in 14.1. (Select the appropriate column " θ_b calculation").

Pollution factor π_p depending on the operating cycle rate, voltage V_t and current I_t (transient state)

Cycle rate: N operating cycles per hour	F_y
$N < 0.1$	0,1
$0,1 \leq N < 1$	N
$N \geq 1$	1

The voltage V_t and the current I_t values (transient state) are given in 14.1. (Select the appropriate column " π_p calculation")

F_y

V_t, I_t

Influence factors:	π_p			
• $V_t \cdot I_t \cdot F_y$ (product) • gaseous environment • relay type	$V_t \cdot I_t \cdot F_y \leq 10$			$V_t \cdot I_t \cdot F_y \geq 10$
	pollution level			Any environment
	Low*	Moderate*	High*	
Hermetically sealed	2	2	2	1
Sealed	2,5	3	10	1
Dust protected	3	5	20	1
Exposed contact relay	5	10	40	1
See Table 7.				

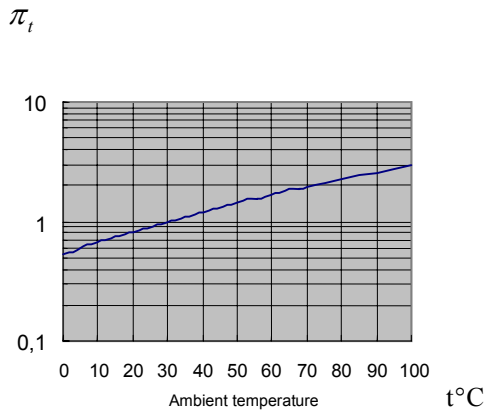
See Table 7.

MATHEMATICAL MODEL

$$\lambda = 1.5 \times \pi_t \times \pi_T \times \pi_p \times \pi_Y \times \pi_C \cdot \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

with $\pi_C = \pi_{C1}$ or π_{C2}

CAUTION !
Life expectancy is
limited !

**Mathematical formula for π_t**

$$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)} \text{ with } t_A : \text{ambient temperature}$$

Necessary information	for
Current, voltage	θ
Load circuit	θ
Number of active contacts	π_C
Operating cycles per hours: mean number N	π_Y, F_Y
Ambient temperature t_A	π_t
Relay type	π_T
Environment	π_p

Number of breaker active contacts "break contact" or "make contact" (form A and B)	π_{C1}
No contact of this type	0
1	1
2	1,5
3	2
4	2,5

Relay type	π_T
Electromechanical (miniature and European)	1
Thermal relay	10

Number of inverter active contact "break contact" and "make contact" (form C)	π_{C2}
No contact of this type	0
1	1,8
2	3
3	4,3
4	5,5
6	8

Operating cycles per hours: mean number N	π_Y
$N \leq 10$	1
$N > 10$	$\sqrt{\frac{N}{10}}$

Failure modes	
Short-circuits	20 %
Open-circuits	80 %

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ac})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

14.5 Industrial relays, high voltage vacuum relays, power mercury wetted relays (IEC 60255)

LIFE EXPECTANCY θ :

$$\theta = \theta_b \frac{1}{\pi_t \times N} \text{ in h}$$

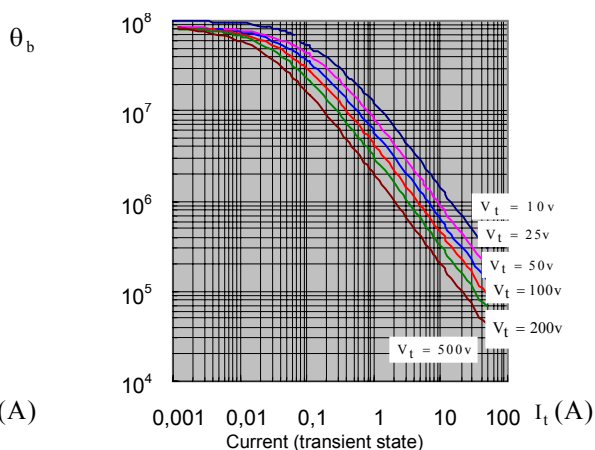
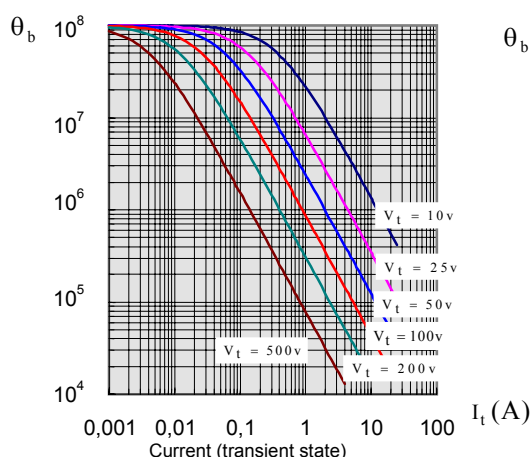
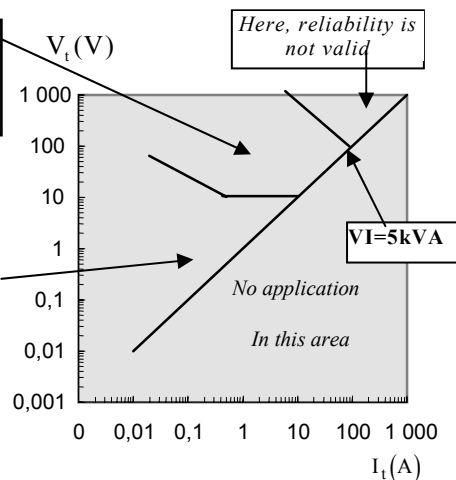
- π_t is given on the next page (same factor as λ for the failure rate).
- N: Number of cycles per hour

Mathematical formulas for θ_b

D.C. relays	A.C. relays
$\theta_b = \frac{10^8}{1 + 100 \left(\frac{V_t}{50} \right)^{1.5} \times \left(\frac{I_t}{2} \right)^{1.3}}$	$\theta_b = \frac{10^8}{1 + 100 \left(\frac{V_t}{220} \right)^{0.5} \times \frac{I_t}{3}}$

In this area, failure rate is assumed to be constant but only within the life expectancy θ .

In this area, failure rate is assumed to be constant during unlimited time



θ_b depending on the voltage V_t and the current I_t in transient state (see clause 14), for d.c. current relays (on the left) and a.c. current relays (on the right).

The voltage V_t and the current I_t values (transient state) are given in clause 14 (select the appropriate column " θ_b calculation")

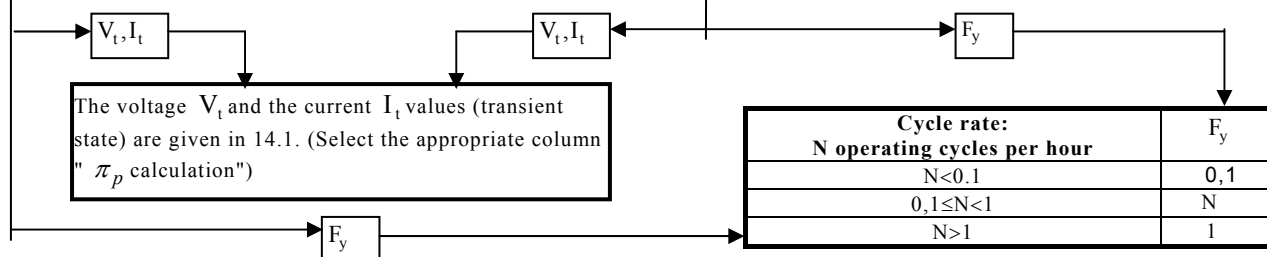
Pollution factor π_p depending on operating cycle rate, gaseous environment and voltage V_t , current I_t (transient state).

Influence factors:	π_p for d.c. relays			
	$V_t \cdot I_t \cdot F_y \leq 50$			$V_t \cdot I_t \cdot F_y > 50$
	Pollution level			Any environment
	Low*	Moderate*	High*	
Mercury wetted relays	1	1	1	1
Hermetically sealed	2	2	2	1
Sealed	2,5	3	10	1
Dust protected	3	5	20	1
Exposed contact relay	5	10	40	1

See Table 7

Influence factors:	π_p for A.C. relays			
	$V_t \cdot I_t \cdot F_y \leq 300$			$V_t \cdot I_t \cdot F_y > 300$
	Pollution level			Any environment
	Low*	Moderate*	High*	
Mercury wetted relays	1	1	1	1
Hermetically sealed	2	2	2	1
Sealed	2,5	3	10	1
Dust protected	3	5	20	1
Exposed contact relay	5	10	40	1

See Table 7

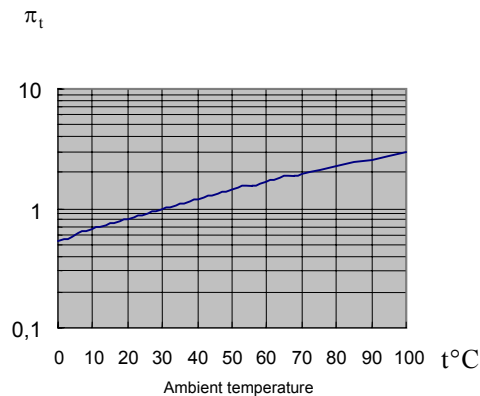


MATHEMATICAL MODEL

$$\lambda = 0.5 \times \pi_t \times \pi_T \times \pi_p \times \pi_Y \times \pi_C \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

with $\pi_C = \pi_{C1}$ or π_{C2}

CAUTION !
Life expectancy is limited !



Necessary information	For
Current, voltage	θ
Load circuit	θ
Number of active contacts	π_C
Operating cycles per hours: mean number N	π_Y, F_Y
Ambient temperature t_A	π_t
Relay type	π_T
Environment	π_p

Mathematical formula for π_t
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

Number of breaker active contacts "break contact" or "make contact" (form A and B)	π_{C1}
No contact of this type	0
1	1
2	1,5
3	2
4	2,5

Relay types	π_T
Industrial relay	2,5
High voltage vacuum relays	2,5
Power mercury wetted relays	1

Number of inverter active contact "break contact" and "make contact" (form C)	π_{C2}
No contact of this type	0
1	1,8
2	3
3	4,3
4	5,5
6	8

Operating cycles per hours: mean number N	π_Y
$N \leq 10$	1
$N > 10$	$\sqrt{\frac{N}{10}}$

Mathematical expression of the	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
Influence factor $(\pi_n)_i$	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{ae})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ae}) variation, during the i^{th} phase of the mission profile	

Failure modes	
Short-circuits	20 %
Open-circuits	80 %

15 Switches and keyboards (IEC 60948)

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times N \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / \text{h}$$

CAUTION!

Life expectancy is limited !

Base failure rate in FIT	λ_0
Toggle switches Push button switches Keyboards	1,5
Rotary switches	2,5

Necessary information	For
Switch type	λ_0, N
Number of contacts	N
Environment	π_E

Type	N
Reversible (toggle or push button)	N = 2x (number of "break-make" contacts)
Others commutators	N = number of contacts
Keyboards	N = number of keys

Life expectancy limitation
Failure rate is assumed to be constant but only within the specified number of switching cycles. For example:
- Toggle and push button switches: between 20 000 and 100 000 (sensitive switches) - Rotary switches: 20 000 - Keyboards: between 500 000 and 1 000 000

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{dc})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

16 Connectors

16.1 Circular, rectangular

16.2 Coaxial connectors

16.3 Connectors for PCBs and related sockets

The failure rate is given for a mated pair of connectors

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \pi_i \times \pi_C \times \pi_M \times \pi_t \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / \text{h}$$

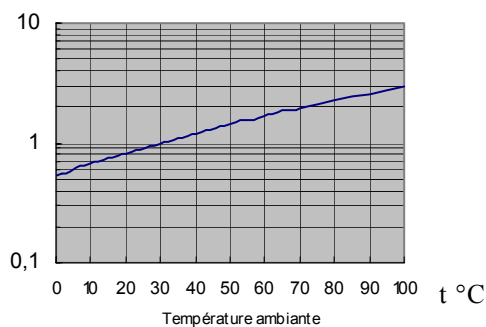
CAUTION !

The life expectancy is limited: The number of mating / unmating cycles must not exceed the specified value (or the value given by the manufacturer).

Base failure rate FIT	λ_0
(these values are given for a contact pair)	
Circular and rectangular connectors	0,5
Coaxial connectors	0,7
Printed circuit board connectors (and comparable)	1
Sockets	1

Necessary information	For
Connector type	λ_0
Contact area material	π_M
Number of active contacts	π_C
Contact current intensity	π_i
Nominal current	π_i
Ambient temperature	π_t

π_t



Contact surface coating (for a mating pair of connectors)	π_M
Gold/Gold	1
Silver/Silver	2
Tin/Tin	3
Others (edge card connector)	8

Contact current intensity	π_i
Ratio: $\frac{\text{Current}}{\text{Nominal current}}$	
≤ 0.5	1
> 0.5	2

Mathematical formula for π_t
$\pi_t = e^{1740 \left(\frac{1}{303} - \frac{1}{273+t_A} \right)}$ with t_A : ambient temperature

Number of active contacts: N	π_C
Coaxial connectors	1
Others	$\sqrt{N}$

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{dc})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

17 Displays, solid state lamps

17.1 Displays (IEC 61747)

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \left(1 + 2.5 \times 10^{-2} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / \text{h}$$

NECESSARY INFORMATION:

Display type

Display types: values in FIT	λ_0
LCD ≤ 10 characters	50
CRT display (10 inches with control electronic)	2 500
LCD display (10 inches with control electronic)	1 900

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{dc})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

17.2 Solid state lamps (IEC 60747)

MATHEMATICAL MODEL

$$\lambda = 2 \times \left(1 + 2.7 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / \text{h}$$

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{dc})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	

18 Protection devices (IEC 60099, IEC 60269, IEC 60738, IEC 61051)

18.1 Thermistors (PTC)

MATHEMATICAL MODEL

$$\lambda = (\lambda_0 + \pi_I \times \lambda_{EOS}) \times 10^{-9} / h$$

NECESSARY INFORMATION:

Device type
Electrical environment

Device type		λ_0 in FIT	π_I
Diodes	Transient voltage suppressor	See 8.2 and 8.3	1
	Trigger transient voltage suppressor	See 8.2 and 8.3	1
Thermistors (PTC)		5	1
Varistors		1	1
Fuses		10	0
Arrestors	Solid state (100 A - 10/1 000 μ s wave)	100	0
	Gas tube (≥ 5 kA - 8/20 μ s wave)	6 000	0

Electrical environment		λ_{EOS}
		FIT
Computer		10
Telecoms.	switching	15
	transmitting	40
	access, subscriber card	
	Subscriber equipment	
Railways, payphone		100
Civilian avionics (on board calculators)		20
Voltage supply, Converters		40

18.2 Varistors

18.3 Fuses

18.4 Arrestors

19 Energy devices, thermal management devices, disk drive

(IEC 60086, IEC 60285, IEC 60535, IEC 60879, IEC 61436, IEC 61440)

19.1 Primary batteries

19.2 Secondary batteries

19.3 Fans

19.4 Thermoelectric coolers

19.5 Disk drive

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times 10^{-9} / h$$

NECESSARY INFORMATION:

Device type

Device type		λ_0 in FIT
Batteries: primary cells*		20
Batteries: secondary cells	Ni-Cd	100
	Li-Ion	150
Fans for integrated circuits (CPU)*		500
Ball bearing fans*		1 500
Bearing fans*		2 500
Thermo-electric cooler		20
Long duration disk drive		2 800

* Caution: life expectancy of these devices is limited.

19.6 Converters (IEC 60146)

MATHEMATICAL MODEL

$$\lambda = \lambda_0 \times \left(1 + 3 \times 10^{-3} \times \left[\sum_{i=1}^j (\pi_n)_i \times (\Delta T_i)^{0.68} \right] \right) \times 10^{-9} / h$$

NECESSARY INFORMATION:

Converter type

Converter type	λ_0 in FIT
Converters < 10 W	100
Converters between 10 W and 30 W	130

Mathematical expression of the Influence factor $(\pi_n)_i$	$n_i \leq 8760$ Cycles/year	$(\pi_n)_i = n_i^{0.76}$
	$n_i > 8760$ Cycles/year	$(\pi_n)_i = 1.7 \times n_i^{0.60}$
n_i : Annual number of cycles with the amplitude ΔT_i		
For an on/off phase	$\Delta T_i = (t_{ac})_i - (t_{dc})_i$	
For a permanent working phase, storage or dormant	ΔT_i = average per cycle of the (t_{ac}) variation, during the i^{th} phase of the mission profile	



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- purchasing agent ☐
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researcher ☐
design engineer ☐
safety engineer ☐
testing engineer ☐
marketing specialist ☐
other.....

Q3 I work for/in/as a:
(tick all that apply)

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consultant ☐
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other.....

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thesis ☐
manufacturing ☐
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(tick one)

- not at all ☐
nearly ☐
fairly well ☐
exactly ☐

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standard is too superficial ☐
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other

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